No detectable edge: cointegration pairs trading on the Russell 3000, 2023–2025

The 500 most liquid point-in-time members of the Russell 3000, traded as same-industry cointegrated pairs re-formed every 126 trading days, with next-day fills and a per-stock cost model, July 2023 to December 2025.

Navnoor Bawa · Note of 21 October 2026 · v33 development run of 21 October 2026, commit 7442705

Code and data github.com/NavnoorBawa/russell3000-pairs-trading
Research-note site navnoorbawa.github.io/russell3000-pairs-trading

Summary

NO DETECTABLE EDGE

No detectable edge, before or after costs. From July 2023 to December 2025 the v33 design returned −0.39% net (Sharpe −0.06, 95% interval −1.29 to +1.16), indistinguishable from zero, from the Gatev distance method (difference +0.73, p = 0.364) and from random pairs (37th percentile). The same rule started on 14 February 2023, the walk-forward, lost money: Sharpe −1.04 (−2.09 to +0.05), one-sided 95% upper bound −0.15. Before costs the trades made $1.52M (Sharpe +0.29, Newey–West p = 0.645), 0.80× the modelled costs: +13.5 bp a trade weighted by size, −7.9 bp equal-weighted, against 16.9 bp of cost. The gross return cannot be told apart from zero or from the cost.

After formation the same rule on the same pairs earns −12.6 bp a trade, against +64.7 bp on the data that selected them; the spreads reach their targets about as often as a random walk's (72 against 65.9, not the 121.0 the selection assumed); and 6 of 115 traded pairs stay cointegrated. A second family, Avellaneda–Lee residual reversion, agrees (Sharpe −0.28, −1.47 to +0.93). The sample rules out a Sharpe ratio above 0.92 (one-sided 95% bound) but cannot tell 0 from 0.5. The pre-registered falsification test on 2026 data, with 16% power against a true Sharpe ratio of 1, is sealed, not prospective: the project was developed during its test period, and its pre-registration is to be filed before any 2026 daily price is fetched.

MAIN BACKTEST · NET SHARPE

−0.06

95% interval −1.29 to +1.16 · Newey–West p = 0.923 · 628 days

WALK-FORWARD (NO REFIT) · NET SHARPE

−1.04

95% interval −2.09 to +0.05 · one-sided 95% bound −0.15 · 716 days

BEFORE COSTS · SHARPE

+0.29

Newey–West p = 0.645 · +13.5 bp a trade size-weighted, −7.9 bp equal-weighted · net breaks even at 0.80× costs

UPPER BOUND ON THE SHARPE RATIO

0.92

One-sided 95%, allowing for fat tails and autocorrelation · 80% power only at 1.47

SECOND FAMILY · NET SHARPE

−0.28

Avellaneda–Lee · 95% interval −1.47 to +0.93 · 10,916 trades

2026 HOLDOUT · NET SHARPE

Sealed

Pre-registration to be filed · sealed, not prospective · 80% power only at a Sharpe of 2.96 over 200 days

1 Introduction

1.1 The question

Pairs trading buys one stock and sells another when the spread between them is unusually wide, and closes the position when it narrows. Gatev et al. (2006) documented excess returns from a simple distance rule on US stocks over four decades; later studies find that the edge has shrunk and is small after realistic costs (Do and Faff 2010; 2012; Rad et al. 2016), and Krauss (2017) surveys the methods. This note asks one question and is built to answer it honestly: **does a cointegration pairs-trading strategy on the Russell 3000 have an edge after realistic execution and costs, out of sample?**

It reports the v33 design. Pairs are selected within one Fama–French 49 industry from the point-in-time members of the Russell 3000, with the Engle–Granger test (Engle and Granger 1987) on formation data only, and the whole selection is re-run every 126 trading days on the trailing three years. Each pair's hedge ratio is tracked with a two-state Kalman filter (Kalman 1960), a z-score of the hedged spread is traded with fills at the next day's close, and every fill pays costs computed from the stock's own liquidity. The design is evaluated in a main backtest from 3 July 2023 to 31 December 2025, in a walk-forward that runs the same rule from 14 February 2023, against two baselines run on the same footing (the distance method of Gatev et al. (2006) and 200 portfolios of random pairs) and beside a second strategy family. The v32 study published on 18 October 2026 is the history this design starts from (Section 10): it is kept as the record, and runs frozen and untested as the third arm of the holdout.

The answer for 2023–2025 is a null. A null is informative only to the extent that the test had power and the measurement was right, so most of this note is about what the sample can rule out (Section 5), why no edge is detected (Section 6) and the checks that the code measures what it claims (Section 3.8). The development period cannot test design choices made on it; a 2026 holdout, sealed at the date of this note, is the confirmatory test, with little power against a modest edge (Section 11).

1.2 What is new

The study uses free data only. Academic studies avoid survivorship with paid point-in-time databases; a backtest on free data usually starts from today's index members instead, and is exposed to the other familiar weaknesses of backtests: flat cost assumptions, significance without power, a null left unexplained, and an untold number of designs tried. Each item below addresses one of these; the last adds a test the development data cannot provide.

- **A point-in-time universe from free SEC filings.** The complete holdings of the iShares Russell 3000 ETF (IWV) are public on Form N-PORT for the last month of each fiscal quarter; 26 quarterly snapshots give index membership as it was at each formation date, without a paid constituent history. The survivorship gap is measured at every formation: 3.33% of index weight at the first, 1.08% at the last (Section 2).
- **Costs from each stock's own liquidity, stressed in full re-runs.** Every fill pays commission, half the quoted spread for the stock's dollar-volume

bucket and square-root market impact on the stock's own volatility and volume, at the next day's close. Each cost and capacity scenario declared in advance re-runs the whole backtest (Section 7).

- **An anatomy of the null.** The same rule in sample and after formation, the persistence of cointegration, the race between targets and stops against a random-walk spread, and the traded pairs' half-lives explain the null trade by trade (Section 6).
- **Power and hurdles.** Each null is reported with the Sharpe ratios it excludes, the Sharpe ratio it could have detected, the track record a given Sharpe ratio would need and the hurdles a result must clear after 32 recorded versions (Section 5).
- **A second strategy family.** Avellaneda–Lee residual reversion, with the paper's parameters, on the same universe rule, fills and costs (Section 9).
- **A record of every trial.** Every run whose result was seen is in an append-only registry, an unpublished exploratory run included, and a variant family declared before any variant ran feeds the data-snooping tests (Section 5.3).
- **A sealed 2026 holdout.** Hypotheses, test, multiplicity correction and decision rule are to be fixed in a pre-registration before any 2026 daily price series is fetched; each arm then runs once and its result is published whatever it is. The test period has already ended, so the test is sealed, not prospective (Section 11).

1.3 Related work

The published record points one way: the classic pairs edge was large before the 2000s, has shrunk since, and is small after realistic costs (Table 1). The closest analogue of this study is Rad et al. (2016), who compare distance, cointegration and copula methods on US stocks; this study differs in period, universe, cost model and inference. Clegg (2014) finds little evidence that pairs cointegrated in one year remain cointegrated in the next, the result this study's anatomy replicates (Exhibit 18). Avellaneda and Lee (2010) trade residuals of single stocks against sector ETFs and factors, the second strategy family of this study (Section 9).

TABLE 1
Published results for pairs trading and residual reversion, as reported by their authors

Study	Sample	Method	Reported result
Gatev et al. (2006)	US, 1962–2002	Distance method, top-20 pairs	Annualised excess returns of up to 11% before costs, falling in later years. The baseline replicated here.
Do and Faff (2010)	US, to 2009	Distance method	Profitability declines over time; the strategy does best in prolonged turbulence.
Avellaneda and Lee (2010)	US, 1997–2007	PCA and sector-ETF residual reversion	Net Sharpe 1.44 (PCA) and 1.1 (ETF) over 1997–2007; PCA only 0.9 over 2003–2007. The second family tested here.
Do and Faff (2012)	US, 1963–2009	Distance method with trading costs	About 30 bp a month for well-matched pairs in refined industries after costs; largely unprofitable after 2002.
Clegg (2014)	US	Persistence of cointegration	Little evidence that pairs cointegrated in one year stay cointegrated the next. Replicated here.
Rad et al. (2016)	US, 1962–2014	Distance, cointegration and copula methods	91, 85 and 43 bp a month before costs; 38, 33 and 5 bp after. From 2009 opportunities for the distance and cointegration methods fall, while the copula method's stay stable. The closest published analogue of this study.
Krauss (2017)	Review	Survey of pairs-trading approaches	Distance, cointegration, time-series, stochastic-control and other approaches compared; open problems include costs and robustness.

Source The studies cited.

2 Data

Every input is free: no account, API key or licence is needed to rebuild the study (Table 2). Prices are not redistributed with the code, under the vendor's terms; each run's fetch manifest lists the symbols and dates a re-run fetches.

TABLE 2
Data sources, all free

Source	What it provides	Used for
Yahoo Finance chart API	Daily open, high, low, close and volume, splits and dividends	Prices of the universe, the 11 sector SPDR ETFs and the VIX
SEC EDGAR: Form N-PORT filings of IWV	Quarterly holdings of the iShares Russell 3000 ETF, with each holding's value per share	Point-in-time index membership; the anchor of point-in-time price levels
OpenFIGI mapping API	CUSIP to ticker	Joining holdings to price histories
SEC EDGAR: company submissions	SIC industry code per registrant	Fama–French 49 industries for same-industry pairs; sector ETFs for the second family
Kenneth French data library	Fama–French five factors, momentum, short- and long-term reversal, one-month T-bill	Factor attribution
FRED	Three-month Treasury bill rate	The cash comparison
Nasdaq stock screener	Market capitalisation of US-listed common stocks	The v32 universe and the frozen arm S3

2.1 Point-in-time membership from N-PORT filings

A registered fund reports its complete portfolio holdings on Form N-PORT, and the report for the third month of each fiscal quarter is made public on EDGAR. IWV tracks the Russell 3000, so its filings give index membership as it was, quarter by quarter, at no cost. The study reads the equity-common holdings of 26 snapshots from 30 September 2019 to 31 December 2025, 2,581 to 2,895 holdings each, and maps each holding's CUSIP (or ISIN) to a ticker through OpenFIGI, N-PORT's own reported tickers, issuer successions and a check against N-PORT's value per share (Exhibit 1). A snapshot is in force from the date its holdings describe until the next one, so it is used about two months before its filing was public: IWV's holdings stand in for the index membership that FTSE Russell had already published. Two caveats: IWV is a fund, not the index, and holds fewer names than the index; and additions and deletions within a quarter appear only at the next quarter end. Snapshots dated 1 January 2026 or later (2 filings) are listed but not downloaded: their holdings carry a market value, and so a 2026 price, for every member.

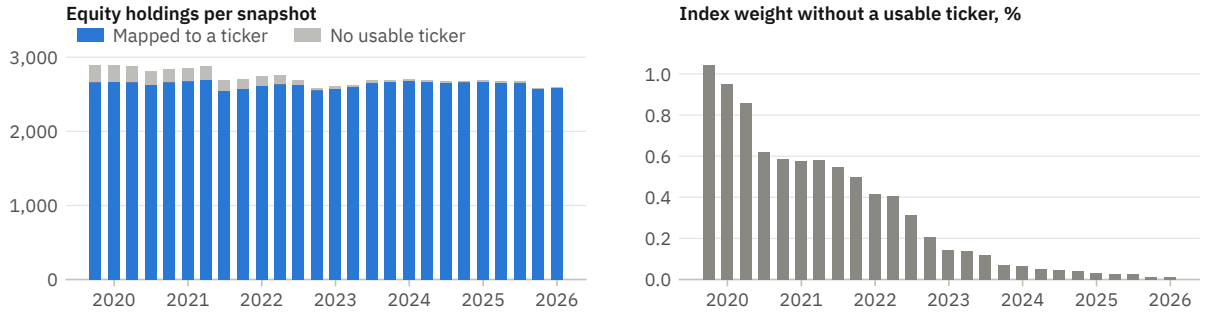
2.2 The universe of the run

At each formation the members of the snapshot in force that have price history go through the formation screen on the trailing window: a point-in-time close of at least $5 and a dollar volume of at least $10M a day on average, annualised volatility between 5% and 80%, and fewer than 10% missing closes. The 500 most liquid by dollar volume enter the pair search, and a pair's two legs must share one Fama–French 49 industry, from each registrant's SEC SIC code. Of the symbols requested, daily prices from 2 January 2020 to 31 December 2025 exist for 82.2% (2,538 symbols); the vendor answered HTTP 404 for 14.2%, symbols it no longer serves, and 3.6% failed the fetch or the data checks. The fetch-coverage rule, which counts 404s and failed data checks as resolved, is met at 98.8% each symbol's outcome is in the run's fetch manifest, `results/v33_dev/price_fetch_manifest.json`.

EXHIBIT 1

Free filings give 26 quarterly snapshots of index membership; the index weight without a usable ticker falls from 1.04% to 0.01%

Equity holdings of the iShares Russell 3000 ETF in its N-PORT filings, 2019–2025, and the share of their value that cannot be mapped to a ticker



Notes A holding without a usable ticker is one no source maps; one whose last ticker now belongs to another issuer; one whose mapping a price check against N-PORT's value per share rejected; or one that would take the ticker another holding of the snapshot has as its own line. The last three are left unmapped on purpose, so that no member is joined to another security's prices. The share is larger in older snapshots, as expected if most such holdings are securities that have since delisted; its weight is part of the survivorship gap.
Source SEC EDGAR, Form N-PORT-P filings of the iShares Russell 3000 ETF (iShares Trust, series S000004341); tickers from OpenFIGI and N-PORT; author's calculations.

2.3 The survivorship gap, measured

The survivorship gap at a formation date is the index weight of members a backtest cannot trade: holdings without a usable ticker, plus members with a ticker but no price history. At the first formation 444 of 2,574 members had no price history (3.192% of index weight) and holdings without a usable ticker added 0.142%; at the last, 237 of 2,658 members had none. By the outcome of each symbol's fetch, 2.75 points of the first formation's missing weight are names the vendor no longer serves (HTTP 404), largely companies acquired or delisted since; 0.34 points failed the fetch and 0.10 the data checks. At the last formation the three are 0.72, 0.25 and 0.09 points. The gap falls from 3.33% to 1.08% (Exhibit 2) because the older the snapshot, the more of its members have since left the market. It is concentrated in large names: among the 500 heaviest members at the first formation, 19 (1.29% of index weight) have no history, 16 of them names the vendor no longer serves, among them Activision Blizzard, Pioneer Natural Resources, VMware, Seagen and Splunk, all acquired since, and SVB Financial, whose bank failed in March 2023, inside the walk-forward. Takeover targets and failures are the names likeliest to have had jump-driven spread breaks in a same-industry pair, so the gap probably flatters a convergence strategy and makes the null more convincing, not less (Section 12). Failed fetches are a data-quality gap, not survivorship.

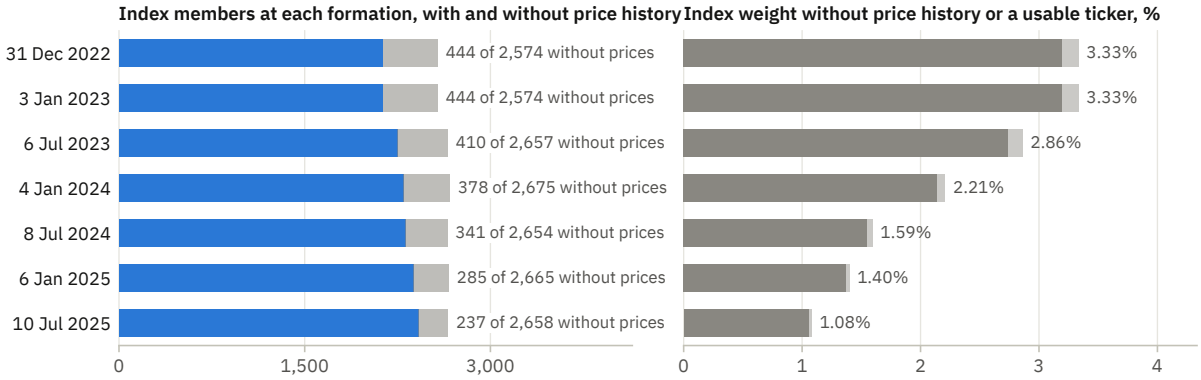
2.4 Price levels anchored to N-PORT

The vendor back-applies every split and dividend up to the fetch date but reports split events only inside the requested window. Early price levels of names that split or paid dividends later are therefore rescaled, and a screen on the price level, such as the $5 floor, or a cost term per share, would read a level the market never showed. v33 anchors the point-in-time close to N-PORT's own value per share at every snapshot date. The anchoring moved 732 of 2,538 price histories (28.8%) by more than 10%, with ratios from 1/80 to 25.6; the largest are splits in 2026 that the vendor had applied to the whole

EXHIBIT 2

The survivorship gap, measured: index weight without price history falls from 3.33% at the first formation to 1.08% at the last

Members of the index snapshot in force at each formation, and the share of index weight the price data do not cover (darker: members without price history; lighter: holdings without a usable ticker)



Notes A pair that includes a member without price history cannot be formed. Shares are of each snapshot's equity value. The first two formations use the same snapshot: the initial selection at the cutoff and the first re-formation, at the first session after it.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, universe by formation; membership from the index fund's holdings in SEC N-PORT filings.

history. Returns, z-scores and hedge ratios are unaffected, since a constant factor cancels in each of them.

2.5 The v32 universe

v32 drew its universe from the Nasdaq stock screener: the 3,000 largest US common stocks by market capitalisation on 18 October 2026, back-filled to 2020. That list holds only survivors, and its ranking uses market values from inside the 2026 test period. It is kept for one purpose: the frozen v32 arm (S3) of the holdout, which is reported but tested in no family for that reason (Section 11).

3 Method

Selection sees formation data only; every trading decision on day t uses data up to day t and fills on day $t + 1$; the evidence is computed after the run (Exhibit 3). Tests change future prices and check that no past decision moves. Table 17 in the appendix states every rule of the design on one page.

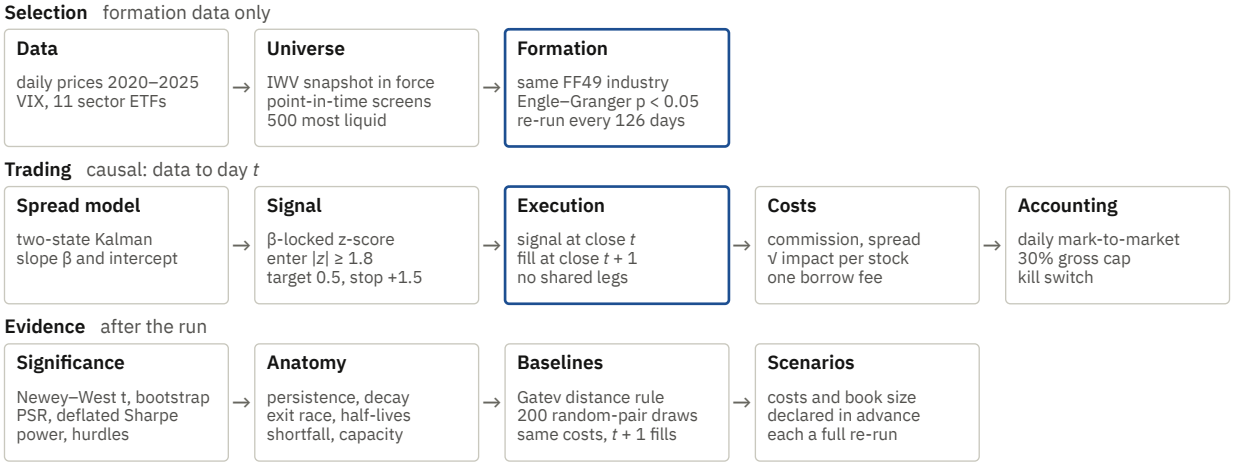
3.1 From v32 to v33

The v33 design differs from the published v32 design in seven fields of the study profile (Table 3). Each change addresses a defect found in v32′s design or in its development run on this same period, so the changes were made after v32′s development results were seen; none re-tunes a trading threshold. The thresholds, the fills and the rest of the cost model are unchanged. Every change was fixed in the profile before the first v33 run. An exploratory v33 run was seen before the published one; both carry the same design hash, so no design field changed after a v33 result was seen, and the exploratory run stays on record (Table 9).

EXHIBIT 3

The pipeline, and what data each stage may see

Stages of one run, grouped by the data each may see

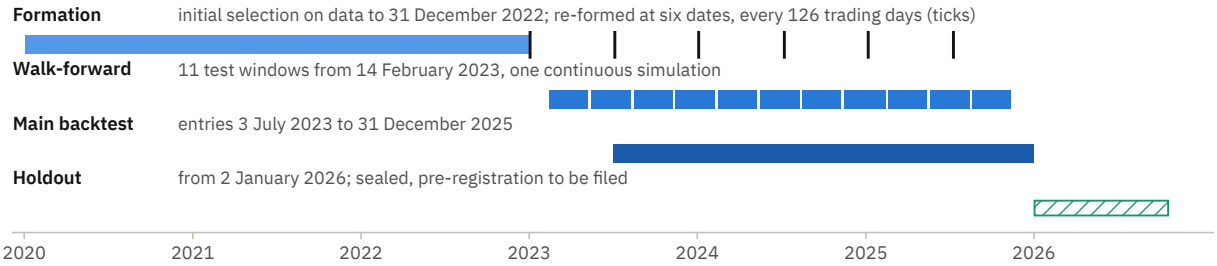


Notes Boxes framed in the accent colour mark the two look-ahead boundaries: the formation cutoff, after which selection sees no data, and the next-day fill, which separates a signal from its trade.
Source Author's design of the v33 study profile.

EXHIBIT 4

Data splits: formation and re-formations, walk-forward, main backtest and the sealed holdout

Calendar 2020–2026



Notes The main backtest and the walk-forward share 621 of the main backtest's 628 days and trade the same pair formations: they are one experiment from two start dates, not two confirmations.
Source Author's study calendar (calendar dev; the holdout is calendar holdout2026).

3.2 Formation

Pairs are first selected on data to 31 December 2022, and the whole selection is re-run every 126 trading days on the trailing three years: 7 formations in the run, of which the first, at the cutoff, was replaced at the first session after it and governed no trade, so 6 governed trading. Among the 500 most liquid screened members, every pair within one Fama–French 49 industry with a positive return correlation of at least 0.25 is tested for cointegration with the two-step Engle–Granger test in both directions, Bonferroni-adjusted across the two, at $p < 0.05$. A pair is kept if the half-life of its cointegrating residual lies between 4 and 25 trading days and a quality score exceeds 0.4: at the first formation, 309 of 5,082 pairs tested (Section 5.4). A formation's pairs trade until the next formation replaces them; positions already open run off.

3.3 Spread model

The hedge ratio is the slope state of a two-state Kalman filter on log prices,

$$\log p_{1,t} = \alpha_t + \beta_t \log p_{2,t} + \varepsilon_t,$$

where (α_t, β_t) follows a random walk, both log prices are centred on their burn-in means and the filter is seeded by ordinary least squares. The inter-

TABLE 3
From v32 to v33: what changed and why

Field	v32, as published	v33	Why
Universe	The 3,000 largest US common stocks by market cap on 18 October 2026, back-filled to 2020	Russell 3000 members as held by the iShares Russell 3000 ETF in its SEC N-PORT filings, the snapshot in force at each formation	Today's largest names are survivors chosen with hindsight; point-in-time membership removes that look-ahead and makes the survivorship gap a measured quantity
Names searched	300 most liquid	500 most liquid	A size at which both strategy families trade the same liquid names
Candidate pairs	Any two names	Two names in the same Fama–French 49 industry	An economic link is the premise of convergence (Do and Faff 2010); fewer tests raise the power of the multiple-testing correction (5,082 against 35,491)
Re-formation	The 2022 pool re-tested every 90 trading days; pairs can drop out, none can enter	The whole selection re-run every 126 trading days on the trailing three years	A pool that can only shrink goes stale, and v32's 126-day re-test had too little power to tell broken pairs from healthy ones; re-forming on a schedule is Gatev's formation-and-trading cycle
Regime gate	VIX and sector-dispersion scaling, with a hard stop on new entries	Off (a declared variant)	Its thresholds were fitted on windows now inside the test period; in the v32 run it blocked new entries on 248 of 628 days
Ranking	A learned classifier orders same-day candidates	Signal strength × pair quality (the learned ranker is a declared variant)	v32's ranker showed no skill on the trades after its training window (AUC 0.514; 0.5 is none)
Borrow and financing	25 bp general-collateral borrow plus a 50 bp prime-broker spread on the short half of gross	One 0.38% fee a year on the short notional	The two charges double-count the cost of shorting; the fee is a median buy-side indicative fee (the execution-cost note gives its source)

Notes Each change was made after v32's development results on the same period were seen, and fixed in the profile before any v33 run. The reasons are stated beside each setting of the study profiles in `pairs_trading/study_config.py`; the execution-cost note, `docs/EXECUTION_COSTS.md`, gives the source of the borrow fee.

cept makes β a hedge ratio rather than a ratio of price levels, and centring makes every signal invariant to a vendor's back-adjustment of history for a later split or dividend, which shifts each log price by a constant.

3.4 Signal and exits

One z-score drives every decision. At a signal on day t_0 the hedge ratio is locked at its current value and the spread is standardised by its mean and standard deviation over the last 30 to 60 bars (three half-lives), both frozen at the signal:

$$s_t = \log p_{1,t} - \beta_{t_0} \log p_{2,t}, \quad z_t = \frac{s_t - \mu_{t_0}}{\sigma_{t_0}}.$$

A pair is entered when $|z| \geq 1.8$, long the spread when z is negative and short when it is positive. It exits at the target once z has reverted past −0.5 for a long spread or +0.5 for a short one, at the stop when z has moved 1.5 further against the position, and at the time stop after max(10, min(⌊2.5 × half-life⌋, 25)) trading days. Entries also require a pair quality of at least 0.55, a 20-day return correlation of at least 0.20 and a hedge ratio in [0.25, 4]; after three consecutive losing trades a pair is paused for 30 calendar days. Same-day candidates are ordered by signal strength times pair quality. There is no regime gate and no learned ranker.

3.5 Execution, accounting and risk limits

A signal computed on the close of day t is filled at the close of day $t + 1$, for entries and exits alike: in practice a market-on-close order in each stock's closing auction, so both legs print together. Share counts are fixed from the signal-day close and the legs are sized \$1 : \$β. Positions are marked to market daily, costs are booked on fill days, and positions still open at the end of the data are closed. The book has \$100M of notional capital. Each position is sized as a share of current equity, 3–10% before risk scaling (2.2% to 10.0% as traded), scaled by the win rate of the last five closed trades and by the

volatility of recent daily returns. Gross exposure is capped at 30% of equity, at most 8 new entries are taken a day, no stock is in two open pairs, and drawdown, daily-loss and exposure kill switches flatten the book. The cap and the exposure kill switch apply to each position's notional as a share of equity at entry; marked to market, gross exposure drifts slightly above 30%. Idle cash earns no interest.

3.6 Walk-forward

v33 has no model to refit, so its walk-forward is the main backtest's rule on the same pair formations, started on 14 February 2023 instead of 3 July 2023 and run as one continuous simulation to 19 December 2025. It is cut into 11 test windows of 63 trading days (the last runs to the walk-forward's last day; Table 19 in the appendix), and positions carry across window boundaries. It is not out of sample for the thresholds, which were chosen across versions on this same data, so this note calls it the walk-forward and reserves "out of sample" for the 2026 holdout. Over the 621 days the two series share, their daily returns correlate at 0.50 and their Sharpe ratios are −0.09 and −0.82: the 30% cap binds, so which signals the book can take depends on what it already holds, and a start four months earlier changes the path.

3.7 Inference

Sharpe ratios are annualised from daily net returns by $\sqrt{252}$. Means are tested with the t-statistic of Newey and West (1987). Intervals for single series are circular block bootstraps with 5,000 resamples and the block length of Politis and White (2004), as corrected by Patton et al. (2009): blocks of 3 days for the main backtest and 4 for the walk-forward. The probabilistic and deflated Sharpe ratios follow Bailey and López de Prado (2012, 2014). Standard errors, power and the one-sided upper bound use the asymptotic variance of the Sharpe estimator for non-normal returns, extended to returns with lag-1 autocorrelation (`pairs_trading/power.py` gives the formula and its source). A one-sided Bonferroni correction allows for the 32 recorded versions; the t of 3.00 that Harvey et al. (2016) propose for new factors and the t of 3.80 of Chordia et al. (2020) are field-wide multiple-testing hurdles, which do not depend on this study's count. The autocorrelation-adjusted annualisation of Lo (2002), which the note does not otherwise use, gives −0.06 for the main backtest and −1.00 for the walk-forward. Differences of Sharpe ratios use the studentized bootstrap of Ledoit and Wolf (2008).

3.8 Validation

A backtest on real data says what a strategy did, not whether the code measures what it claims. A synthetic validation runs the real engine, with its full cost model, on markets whose answer is known, 10 markets per scenario (Table 4, Exhibit 5). Planted cointegrated pairs show a large gross edge; correlated pairs that are not cointegrated, forced into the universe, show none; the Kalman hedge ratio recovers the true slope; and pair selection finds every planted pair while admitting fewer than 5% of unrelated ones. The test suite adds causality checks, each confirmed by re-introducing the bug it guards against: every decision dated on or before T is unchanged when all prices, volumes and macro data after T are replaced; the same trades occur at the

TABLE 4

Known-answer checks on synthetic markets

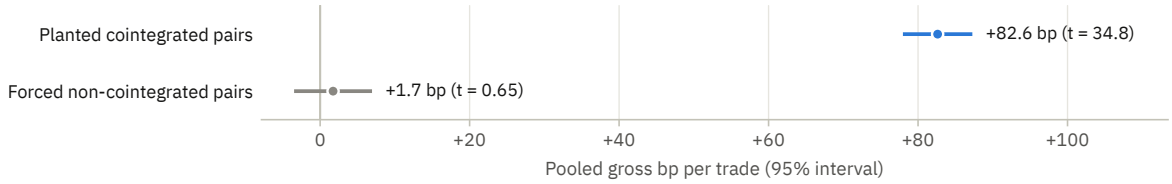
Check	What must happen	Result
Planted cointegrated pairs (OU spreads, half-life 5–14 days)	a large positive gross edge	4,692 trades, +82.6 bp gross per trade (t = 34.8), median net Sharpe 3.47
Correlated, not cointegrated pairs (random-walk spread), forced into the universe	no edge; a positive mean would be look-ahead	5,450 trades, +1.7 bp gross per trade (t = 0.65)
Kalman hedge ratio against the true slope	recover it	median absolute error 0.032 (90th percentile 0.074)
Pair selection with planted pairs among unrelated names	find them; admit at most 5% of unrelated pairs	100% of planted pairs found; 3.3 unrelated pairs admitted per market with planted pairs, 2.6 per market with none; the worst market admitted 2.96% of its unrelated pairs

Notes Pooled over the synthetic markets of each scenario; the edge scenario's net Sharpe ratio is the median across its markets.
Source `results/validation/synthetic_validation.json` (`scripts/synthetic_validation.py`).

EXHIBIT 5

On synthetic markets the engine finds a planted edge (+82.6 bp a trade) and none where there is none (+1.7 bp, t = 0.65)

Pooled gross bp per trade, planted cointegrated pairs against forced non-cointegrated pairs, with 95% intervals



Notes Intervals are ±1.96 standard errors of the pooled gross return per trade across the synthetic markets.
Source `results/validation/synthetic_validation.json` (`scripts/synthetic_validation.py`).

same z-scores when the vendor back-adjusts history for a later split; and the anatomy reconstructs every trade's P&L from prices to the cent. The statistics are tested against reference implementations: statsmodels for Newey–West, arch for the Politis–White block length and the model confidence set, an independent cross-validation for the probability of backtest overfitting, and the authors' code for Ledoit–Wolf and for the Sharpe-ratio power functions.

4 Development-period results

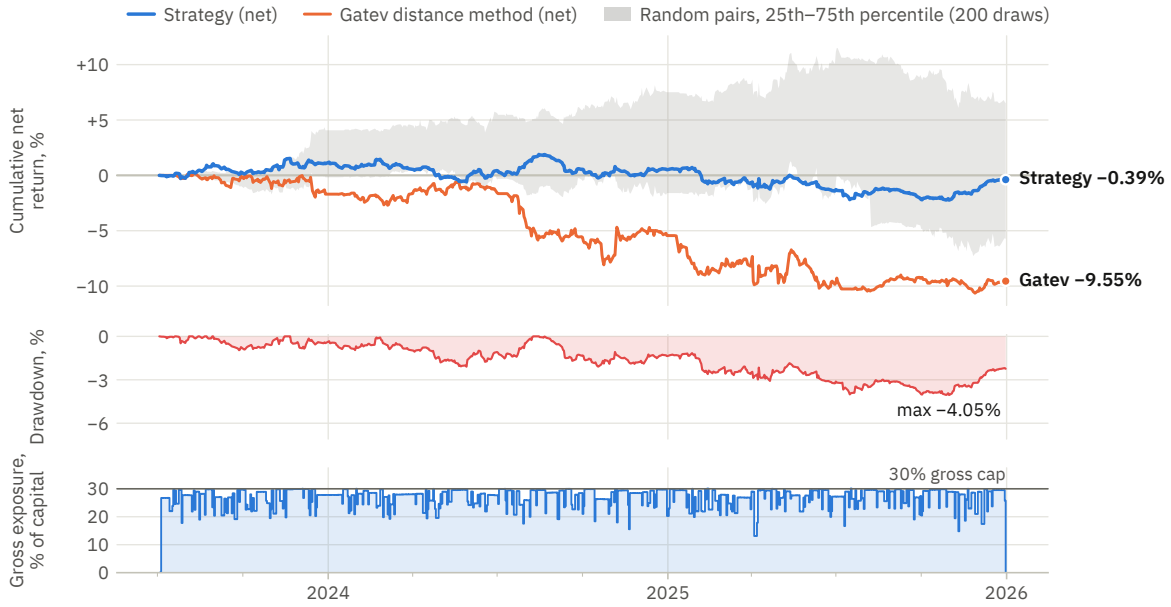
The main backtest enters trades from 3 July 2023 to 31 December 2025 on $100M of notional capital. The start is the development calendar's, which v32 set after the label window of its ranking model; v33 has no ranker but keeps the calendar, so that the two designs are compared on the same dates. Its pairs are formed on the point-in-time index at the end of 2022 and re-formed every 126 trading days on the trailing three years. Its 213 trades returned −0.39%, with a maximum drawdown of −4.05% (Exhibit 6). The book was flat on 2 of 628 days (v32′s book, with its regime gate, on 207) and used 26.9% of capital on average, gross, against a 30% cap. Table 5 sets out the standard statistics of the main backtest, the walk-forward and the Gatev baseline.

The walk-forward points the same way (Exhibit 7). Of its 11 windows, 4 made money and the stitched path ends at −6.24%; the mean window return is −0.58% with a 95% interval from −1.28% to +0.12%. No single month or half-year decides the result (Exhibit 8, Exhibit 9).

EXHIBIT 6

Net of costs the book ends where it started, −0.39% after two and a half years, with the 30% gross cap close to binding throughout

Cumulative net return against the Gatev baseline and random pairs, underwater drawdown and gross exposure; main backtest, daily, July 2023 – December 2025



Notes The baselines' returns are on committed capital (pair P&L per dollar of leg notional, equal-weighted), the strategy's on its whole capital, of which it deploys about a quarter gross: compare their Sharpe ratios (Exhibit 10), which do not depend on that basis, rather than these levels. The band is the middle half, day by day, of 200 random-pair portfolios traded with the distance rule under the same costs; on the last day their 5th and 95th percentiles stand at −92.3% and +17.8%, beyond this scale (an unhedged random pair can lose most of its committed capital). Gross exposure is the run's own daily export: the cap applies to exposure at entry, and marked to market gross exposure drifts slightly above 30% (maximum 30.11%, on 5 days).
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance; baseline: Gatev et al. (2006) distance rule with the same $t + 1$ fills and costs.

TABLE 5

Performance of the main backtest, the walk-forward and the Gatev baseline, net of costs

	Main backtest	Walk-forward	Gatev distance method
Trading days	628	716	628
Total net return, %	−0.39	−6.24	−9.55
Annualised net return, % (95% interval)	−0.13 [−2.82, +2.55]	−2.25 [−4.60, +0.11]	−3.91 [−9.39, +1.58]
Annualised volatility, % (95% interval)	2.15 [1.91, 2.42]	2.16 [1.94, 2.39]	4.91 [4.18, 5.69]
Sharpe ratio (95% interval)	−0.06 [−1.29, +1.16]	−1.04 [−2.09, +0.05]	−0.80 [−1.98, +0.40]
Maximum drawdown, %	−4.05	−6.46	−10.82
Drawdown length, trading days: to trough / to recovery	304 / not regained	707 / not regained	579 / not regained
Average gross exposure, % of capital	26.9	—	—
Net return on average gross exposure, % a year	−0.49	—	—
Trades	213	256	120
Hit rate, % of trades	47.9	44.1	—
Average holding period, trading days	15.1	—	62.5
Annual turnover, × capital	9.0	—	—

Notes Annualised returns are mean daily returns × 252 with a Newey–West 95% interval; volatility intervals are circular block bootstraps with the Sharpe ratio's block length. The Gatev return is per unit of committed per-leg notional, the strategy's per unit of capital; compare Sharpe ratios across columns, not return or volatility levels. A dash marks a statistic the run does not report for that series.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance; baseline: Gatev et al. (2006) distance rule.

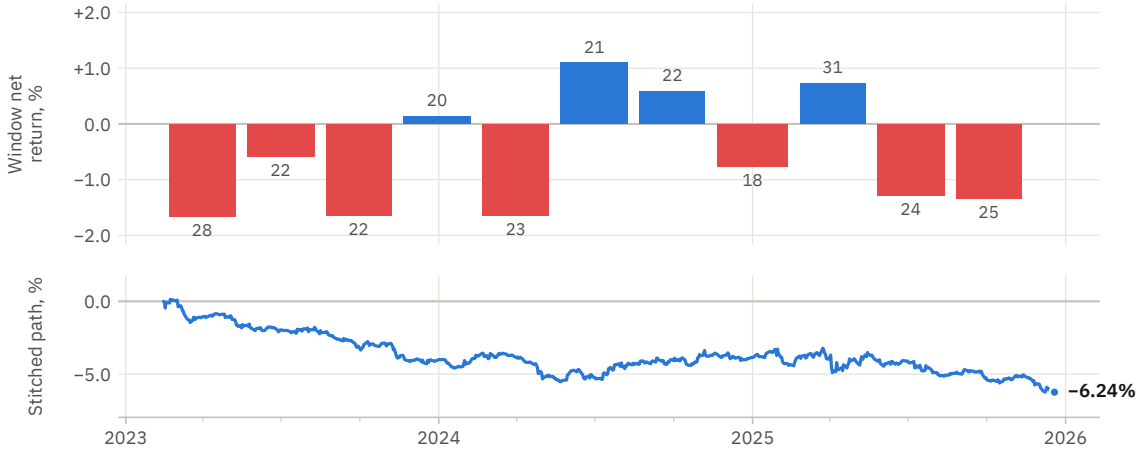
5 Inference

Is the result significant? Not in the strategy's favour: every interval in Exhibit 10 contains zero, gross or net, in both strategy families, and the one test that rejects, the walk-forward's one-sided test, rejects in the direction of a loss; Table 8 gives the tests. The more useful question is what the data can rule out. With 628 daily observations the standard error of the annualised Sharpe ratio is 0.59, allowing for the returns' fat tails (skewness

EXHIBIT 7

Started in February 2023, the same rule made money in 4 of 11 quarterly windows; the stitched path ends at −6.24%

Net return per 63-day test window (bars, labelled with the trades opened) and the compounded daily path; February 2023 – December 2025



Notes The windows are calendar slices of one continuous simulation; positions carry across their boundaries. With 11 windows the test has little power: mean −0.58% per window, 95% interval −1.28% to +0.12%, p = 0.096.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, walk-forward windows and daily path.

EXHIBIT 8

No single month decides the result: 13 of 30 months were positive, and the best, August 2023, earned +1.13%

Monthly net return, % of capital; the year column compounds the months shown

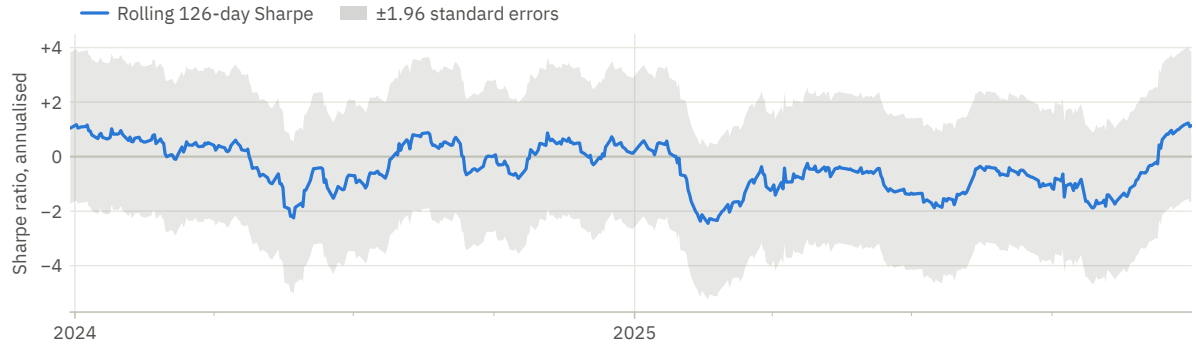
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2023							−0.27	**+1.13**	−0.56	+0.14	+0.55	+0.07	+1.05
2024	−0.22	+0.21	−0.45	−0.63	−0.02	+0.45	+0.66	+0.50	−1.02	−0.28	−0.19	+0.45	−0.55
2025	−0.13	−0.76	−0.36	+0.19	−0.28	−0.59	−0.08	+0.19	−0.79	−0.15	+0.88	+1.00	−0.88

Notes Sharpe ratio by calendar year: +1.04 in 2023 (from July), −0.26 in 2024 and −0.37 in 2025. This grid is the evaluated period: a rule fitted to it would be data snooping.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, monthly performance.

EXHIBIT 9

Half-year Sharpe ratios range from −2.5 to +1.2, and no window's band clears zero

Rolling 126-day Sharpe ratio, annualised, with a ±1.96 standard-error band; main backtest



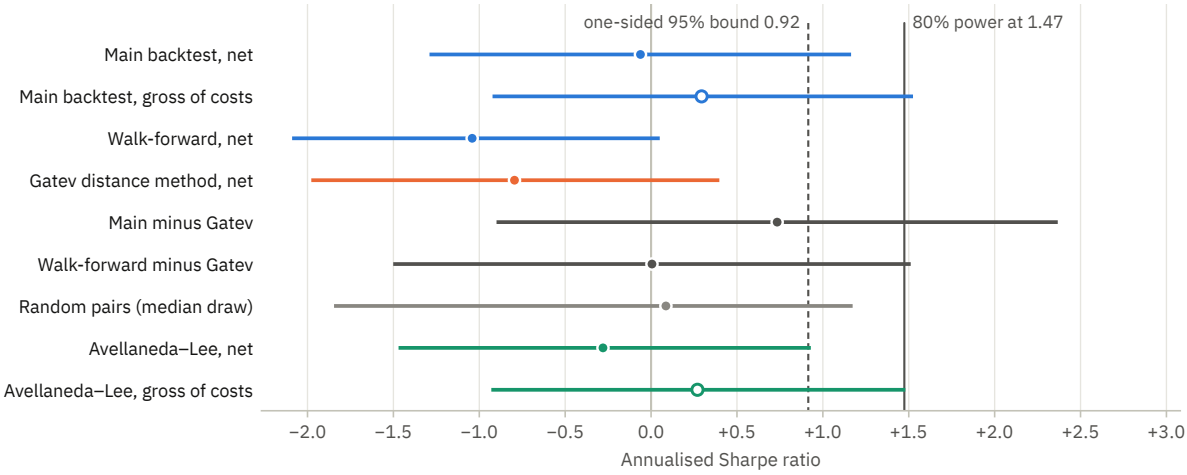
Notes SE = √((1 + SR²/2)/*n*) per day with *n* = 126, annualised, the independent-returns standard error of Lo (2002). 0 of 503 overlapping windows clear zero; overlapping windows are not independent tests.
Source Author's calculation from the daily net returns of the v33 run.

0.23, non-excess kurtosis 7.7) and their lag-1 autocorrelation of −0.065, so the one-sided 95% upper bound is 0.92. A true Sharpe ratio would need to be about 1.47 to be detected with 80% probability. A Sharpe ratio of 1.0 is rejected (p = 0.037); 0.5 is not (p = 0.172). An observed Sharpe ratio of 0.5

EXHIBIT 10

Every bootstrap interval spans zero, gross or net, in both families; the main backtest excludes Sharpe ratios above 0.92, and the walk-forward's one-sided bound is below zero

Annualised Sharpe ratio with its 95% interval; development period, net of costs unless stated (open markers: before costs)



Notes Intervals are circular block bootstraps (Politis–White block length, 5,000 resamples) for single series and Ledoit–Wolf studentized bootstraps for differences; the random-pair row shows the 5th–95th percentile spread of 200 draws, not a confidence interval. The gross-of-cost, Gatev and Avellaneda–Lee intervals use the run's own bootstrap. The bound and the 80%-power line are the main backtest's. The walk-forward's comparison with Gatev uses the 621 days the two series share.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance; Gatev et al. (2006) distance rule; `results/residual_dev`.

TABLE 6

What the development sample rules out

True annual Sharpe x	p-value of H₀: SR ≥ x	Years for an observed SR = x to be significant
0.5	0.172	9.5
1.0	0.037	2.4
1.5	0.004	1.1
2.0	< 0.001	0.6

Notes One-sided, at the observed Sharpe ratio, under the non-normal AR(1) variance. A strategy whose true Sharpe ratio is *x* reaches significance after the stated years only about half the time; 80% power needs about 2.3 times as long (Bailey and López de Prado 2012).

TABLE 7

Sharpe hurdles: Bonferroni over 32 recorded versions, and two field-wide hurdles

Hurdle	t	Annual Sharpe
One test, one-sided 5%	1.64	0.98
Bonferroni over 32 trials, one-sided	—	1.75
Harvey, Liu and Zhu (2016), field-wide	3.00	1.78
Chordia, Goyal and Saretto (2020), field-wide	3.80	2.26

Notes The annual Sharpe ratio at which the main backtest's one-sided t would clear each hurdle at this sample length, under the same variance.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, evidence block.

becomes significant only after about 9.5 years of daily data, and a strategy whose true Sharpe ratio is 0.5 reaches that bar only about half the time; at 1.0 the length is 2.4 years.

5.1 Power, the upper bound and the hurdles

A test that cannot reject zero is evidence of no edge only to the extent that it could have detected one. At this sample length the critical Sharpe ratio of a one-sided 5% test is 0.98 and the Sharpe ratio detected with 80% probability 1.47; the one-sided 95% upper bound, 0.92, is the largest Sharpe ratio the data do not reject (Table 6). The sample therefore excludes a strategy of Sharpe 1.0 or more and says nothing useful about 0.5. Multiple-testing hurdles sit higher still (Table 7): a Sharpe ratio of 1.75 for a one-sided Bonferroni correction over the 32 recorded versions, and the field-wide hurdles of 1.78 (t = 3.00) and 2.26 (t = 3.80), all above the upper bound. The walk-forward's own one-sided upper bound is −0.15: over its 716 days the one-sided test rejects a Sharpe ratio of zero or more in favour of a loss (its two-sided bootstrap interval, −2.09 to +0.05, just includes zero), and a loss is not an edge either.

TABLE 8

Significance tests, v33 development run of 21 October 2026

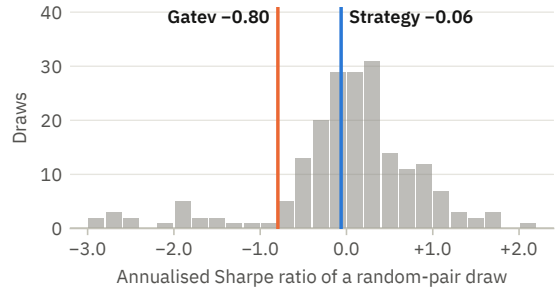
Test	Main backtest	Walk-forward (daily)
Annualised Sharpe ratio, net	−0.06	−1.04
Newey–West t (mean ≠ 0)	−0.10	−1.87
Newey–West p	0.923	0.061
Probabilistic Sharpe ratio (one minus the one-sided p of SR ≤ 0)	46.1%	3.9%
Deflated Sharpe (32 trials)	1.4%	0.0%
Block-bootstrap 95% interval	[−1.29, +1.16]	[−2.09, +0.05]
Sharpe ratio before costs (Newey–West p)	+0.29 (0.645)	−0.69 (0.211)
Minimum detectable Sharpe (80% power, one-sided 5%)	1.47	1.37
One-sided 95% upper bound	0.92	−0.15
Gatev distance method, Sharpe [95% interval]	−0.80 [−1.98, +0.40]	−0.83 (on 621 shared days)
Strategy minus Gatev, Sharpe (Ledoit–Wolf p)	+0.73 (0.364)	+0.01 (0.993)

Notes Newey–West t and p test a zero mean of daily net returns. The probabilistic and deflated Sharpe ratios are one minus a one-sided p-value: of the observed Sharpe ratio against zero, and against the expected maximum Sharpe ratio of the stated number of independent null trials; neither is a probability that the strategy has an edge. Power and the upper bound use the non-normal AR(1) variance. The walk-forward's comparison with Gatev uses the 621 days both series cover, on which the walk-forward's own Sharpe ratio is −0.82; no interval is computed for the Gatev Sharpe ratio on those days.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance; tests of Newey and West (1987), Bailey and López de Prado (2014) and Ledoit and Wolf (2008).

EXHIBIT 11

The strategy sits at the 37th percentile of 200 random-pair portfolios

Annualised Sharpe ratio of random 20-pair draws traded with the distance rule under the same costs

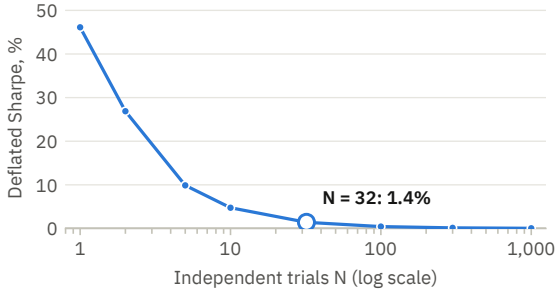


Notes Median draw +0.09, 5th–95th percentile −1.85 to +1.17. The control trades random pairs with the Gatev rule, not this engine: a baseline for the distance method and only a rough one for this strategy.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance.

EXHIBIT 12

Deflated for 32 trials, the Sharpe ratio is far from significant (deflated Sharpe ratio 1.4%)

Deflated Sharpe ratio of the main backtest against the number of independent trials *N*



Notes *N* = 1 is the probabilistic Sharpe ratio; *N* = 32 counts the versions before this one, a lower bound on the designs tried.
Source Author's calculation from the daily net returns.

5.2 Baselines

The Gatev distance method, run on the same universe and period with the same $t+1$ fills and cost model, returned −9.55% (Sharpe −0.80; 95% interval −1.98 to +0.40). Its return is per unit of committed per-leg notional, at 4.91% annualised volatility against the strategy's 2.15% on capital, so the Sharpe ratios, not the returns, are comparable (Table 5). The difference in Sharpe ratios is +0.73 in the main backtest (Ledoit–Wolf p = 0.364) and +0.01 in the walk-forward, on the 621 days the two series share (p = 0.993). Against 200 portfolios of random pairs, traded with the Gatev rule under the same costs, the strategy sits at the 37th percentile (Exhibit 11).

5.3 Deflated Sharpe, data snooping and the trial record

Every design choice in this study was made on 2020–2025 data over 33 versions (Table 21). The deflated Sharpe ratio (Bailey and López de Prado 2014) is one minus the one-sided p-value of the observed Sharpe ratio against the expected maximum Sharpe ratio of N independent null trials; at N = 32, a

TABLE 9
Every trial whose result has been seen

Trial	Design · calendar	Commit	Sharpe	Return, %	Trades	Walk-forward Sharpe
v32_dev	v32 · dev	711b18c	−0.07	−0.30	161	−0.33
v33_dev_explore	v33 · dev	6523ace*	−0.51	−3.23	242	+0.13
rehearsal2025_unscripted	v33 · 2025	82ba529	−1.06	−2.25	97	—
rehearsal2025_unscripted	v32 · 2025	9cf35cf	−1.07	−0.44	12	—
v33_dev	v33 · dev	7442705	−0.06	−0.39	213	−1.04
residual_dev	Avellaneda–Lee · dev	7442705	−0.28	−3.37	10,916	—

Notes In the order registered in the append-only `results/trial_registry.jsonl`. * Run with uncommitted changes. The exploratory run and the published v33 run carry the same design hash; between them the point-in-time universe was reviewed and rebuilt, price levels were anchored to N-PORT, and code the profile does not record changed with them. The residual arm's line holds no run record, so its numbers are those of its published run. The variant family and the holdout will be added as one trial each.
Source `results/trial_registry.jsonl`.

lower bound on the designs tried, it is 1.4% for the main backtest and 0.0% for the walk-forward (Exhibit 12). Harvey et al. (2016) make the same point for the factor literature.

N is not known, and the deflated Sharpe ratio is only as good as the guess. The data-snooping tests replace the guess with a declared family. On 20 October 2026, before any variant ran, the study declared a 27-point grid of the entry, exit and stop thresholds and 11 design toggles around the v33 design (Table 18 in the appendix); each variant runs once on the development calendar, and the design is not re-chosen from the results. The Reality Check of White (2000) and the SPA test of Hansen (2005) ask whether the best variant beats cash; the stepdown of Romano and Wolf (2005) identifies which variants do, with familywise error control; the model confidence set of Hansen et al. (2011) keeps the variants that cannot be told apart from the best; and the probability of backtest overfitting (Bailey et al. 2017) measures how often the in-sample winner falls below the median out of sample. Resampling uses the stationary bootstrap (Politis and Romano 1994). The family's run is in progress; its results are forthcoming (Table 16) and will be reported whatever they show.

Beyond the declared family, every trial whose result has been seen is on record, in the order registered (Table 9). The exploratory v33 run, on the first point-in-time build and before the universe review and the price anchoring, had a walk-forward Sharpe ratio of +0.13; it is not published, and it stays in the record rather than being dropped. The rehearsal runs, on the holdout calendar moved one year back, are development data; no decision depends on them.

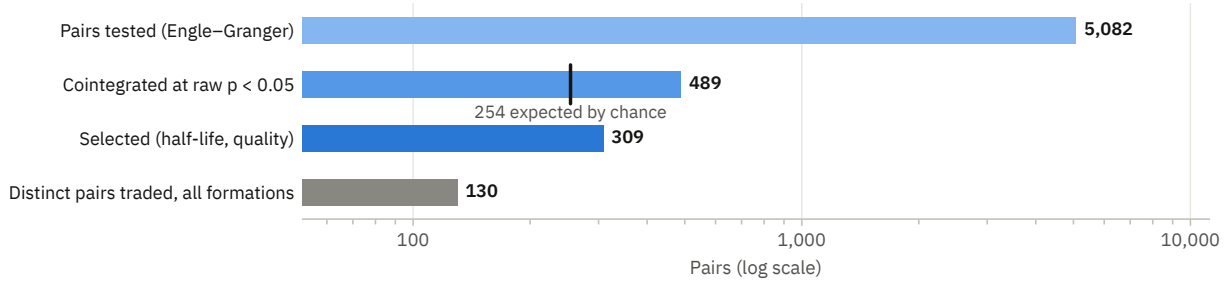
5.4 Multiple testing in pair selection

Pair selection is itself a multiple test. At the first formation 5,082 same-industry pairs were tested and 489 were "cointegrated" at $p < 0.05$, against 254.1 expected by chance alone. That count is an upper bound on chance rejections, since it treats every pair as null and the two-direction Bonferroni p-value is conservative, so up to 52% of the "cointegrated" pairs could be chance. None survives a Benjamini–Hochberg correction (Benjamini and Hochberg 1995) at $q = 0.05$, and 1 at $q = 0.10$: the excess is spread thinly over many weak relationships. Selection kept 309 pairs at that formation, which governed no trade: it was replaced at the first session after the cutoff, before trading began. Across the 6 formations that governed trading, 130 distinct

EXHIBIT 13

From 5,082 pairs tested at the first formation, chance alone could account for up to about half the "cointegrated" ones

Pairs at each selection stage of the first formation (data to 31 December 2022), and, separately, the distinct pairs traded across the 6 formations that governed trading; log scale



Notes The tick marks the count expected by chance, an upper bound. The last bar is not a share of the stage above it: the first formation governed no trade. Pairs share legs, so the tests are dependent; Benjamini–Hochberg is a diagnostic here, not a filter. Selection also requires a half-life of 4–25 days and a quality score. The exports keep these counts, not the p-values themselves, so no p-value histogram is drawn.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, FDR diagnostic and trades.

pairs were traded (Exhibit 13). Restricting pairs to one industry cut the tests from v32′s 35,491.

6 Why no edge is detected

No gross edge is detected. Before costs the 213 trades made $1,520.1k (Sharpe +0.29, Newey–West p = 0.645); modelled costs were $1,907.4k, 67.4% of it market impact, so costs would have to fall to 0.80× their modelled level for net P&L to break even (Exhibit 14). Weighted by size, the trades earned +13.5 bp of notional before costs against 16.9 bp of cost (t = 0.40 for the mean gross dollar P&L); equal-weighted, the average trade earned −7.9 bp (t = −0.25) against 16.1 bp. On either basis the gross return is indistinguishable from zero and from the cost, and $782.5k of the gross is market exposure (Exhibit 15).

Four measurements of the evidence layer are consistent with one mechanism. The rule's edge on the data that selected a pair disappears afterwards (Exhibit 17); after formation the spreads move like random walks between the target and the stop (Exhibit 16), with half-lives longer than the rule allows (Exhibit 19); and the traded pairs, like pairs on a broader universe, stay cointegrated only at about the base rate of correlated pairs (Exhibit 18). The selection, in other words, mostly finds pairs whose spreads happened to revert in the formation window. "After formation" is development data: this note keeps "out of sample" for the 2026 holdout.

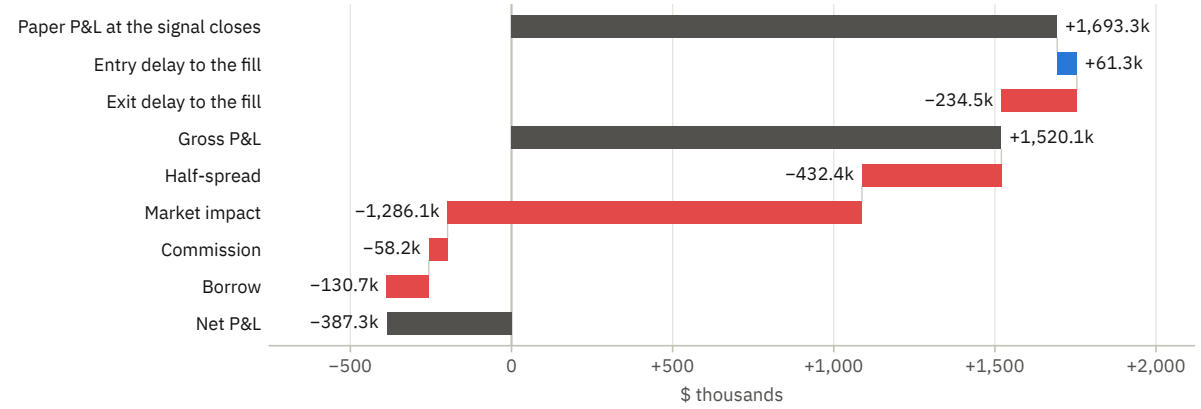
7 Execution and costs

Every fill pays its own costs, from the stock's own liquidity on that day, and fills at the next day's close: a signal seen at the close of day t trades in the closing auction of day $t + 1$. The average trade held 15.1 trading days, and its larger leg was 1.61% of that stock's full-day dollar ADV; the closing auction trades only a fraction of a day's volume, so participation in the auction itself is higher. Annual turnover was 9.0× capital. The cost model was fixed before the run (Table 10), and its uncertain parts are stressed in scenarios declared in

EXHIBIT 14

From paper P&L to net: the next-day fill gives up part of the paper P&L, and costs exceed what the trades earned before them

Implementation shortfall of the main backtest, $ thousands: paper P&L at the signal closes, the delay to the next-day fills, gross P&L, each cost, net P&L

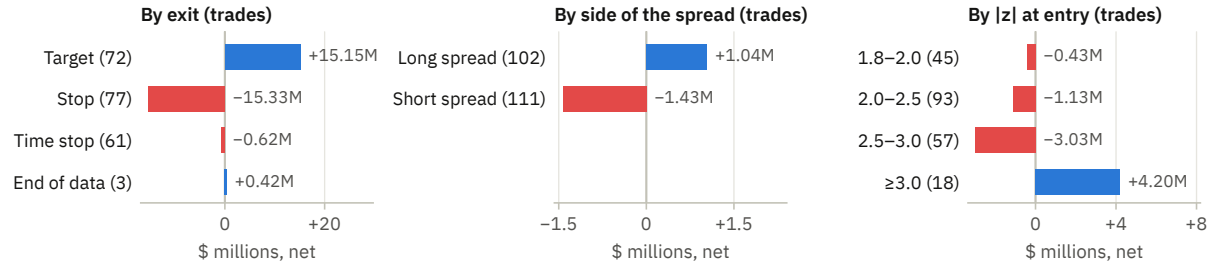


Notes Paper P&L at the signal closes is unattainable, because a signal is known only at the close. Filling at the next close gained $61.3k on entries and lost $234.5k on exits, leaving 13.49 bp of traded notional, weighted by size. The bars close exactly on the change in the book's value: gross 152.0 bp of starting capital, costs 190.7 bp, net −38.7 bp. v33 charges no financing beyond the borrow fee. The same $100M in three-month Treasury bills over the same dates would have earned 12.72%; the backtest's capital earns no interest.
Source `results/v33_dev/anatomy.json` (shortfall, cash; FRED three-month bill rate), reconstructed from the trades and prices and checked against the export to the cent.

EXHIBIT 15

Targets and stops offset each other; long-spread trades made $1.04M and short-spread trades lost $1.43M, consistent with hedge ratios below one

Net P&L of the 213 trades by exit reason, side of the spread and |z| at entry, $ millions; each panel on its own scale



Notes Stops lost $15.33M and targets earned $15.15M, by construction of the two exits. Long-spread trades made $1.04M and short-spread trades lost $1.43M, consistent with the market exposure that hedge ratios below one leave (mean 0.70; 87.8% of trades below one): per dollar of gross, a long-spread trade carries a market beta of +0.26 and a short-spread trade −0.21. The market component accounts for $782.5k of the $1,520.1k gross; without it the gross return per trade is −11.2 bp (t = −0.39). Win rate 47.9% (102 of 213; 95% interval 41.3–54.6%). The |z| buckets describe the past and are not a selection rule.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, trades; `results/v33_dev/anatomy.json`, exposure.

advance; the study's execution-cost note compares the model with published impact estimates at each participation rate (Section 14).

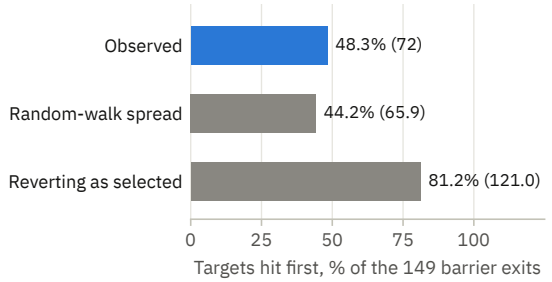
7.1 Cost and capacity scenarios

How much the conclusion depends on the cost model is measured, not asserted. Each scenario declared in advance re-runs the whole backtest (Exhibit 20, Table 11), and every re-run loses money. The re-runs do not hold the trades fixed: trade counts range from 213 to 273. The code shows three reasons. Costs enter the equity path, and every position is sized as a share of current equity, scaled by the win rate of the last five closed trades and by the volatility of recent daily returns. An entry is admitted only while gross exposure stays under the 30% cap, which binds: the headline run averaged 26.9% and the cap turned away 7,866 signals at signal time and 2,261 more at the fill. And a pair is paused for 30 days after three consecutive losing trades

EXHIBIT 16

After formation, spreads hit their targets about as often as a random walk would, not as the selection implies

Targets hit first among the 149 trades that reached the target or the stop, against two nulls (expected count in brackets)

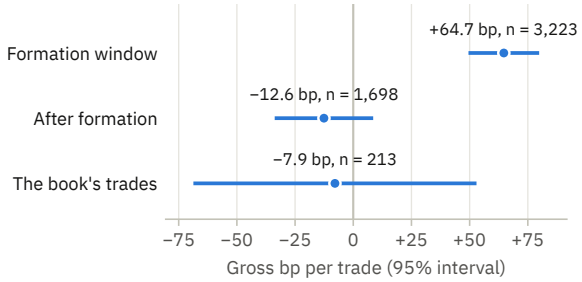


Notes 72 targets and 77 stops. A driftless random-walk spread with each trade's own volatility, entry *z* and time stop expects 65.9 targets (z = +1.02, p = 0.309); a spread reverting at its pair's formation half-life (median 15.8 days) expects 121.0 (z = −10.73). 64 trades ended at the time stop or the end of data.
Source `results/v33_dev/anatomy.json` (exit race; simulated paths per trade).

EXHIBIT 17

The rule's edge on its own pairs is in-sample only: +64.7 bp a trade on the selection data, −12.6 bp after formation

Gross bp per trade of the same rule on the 130 traded pairs, with 95% intervals

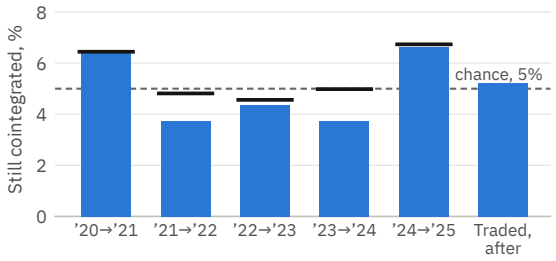


Notes In sample, on each pair's formation window: t = 8.35 (5.60 clustered by month). After formation: t = −1.17. The book's own trades: t = −0.25. Selection on in-sample fit inflates the in-sample edge even when mean reversion is real, but a genuine selection would keep a positive edge after formation; here the post-formation mean is −12.6 bp (95% interval −33.8 to +8.6).
Source `results/v33_dev/anatomy.json` (decay).

EXHIBIT 18

Cointegration persists only at the base rate: a pair cointegrated one year is cointegrated the next about as often as any correlated pair

Share of pairs cointegrated in a year (p < 0.05) that are cointegrated the next: the 300 most liquid names at the end of 2022 (any industry, every pair correlated at 0.25 or more, one-year windows), years 2020–2025, and the traded pairs after formation

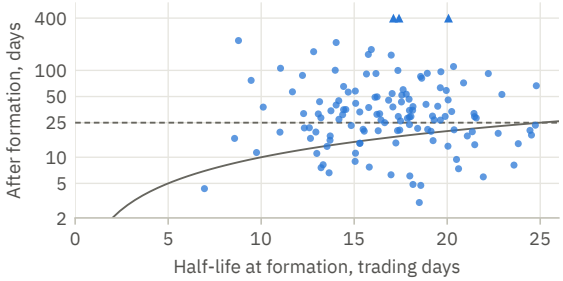


Notes Ticks: the next year's base rate; dashed line: 5%. Pooled over the five pairs of years, 370 of 7,334 stayed cointegrated (5.04%, 95% interval 4.56–5.57%), against 403.4 at the next year's base rate (z = −1.71); of the 115 traded pairs testable after formation, 6 (5.2%). The yearly bars use a broader universe than v33's (500 names, same-industry pairs, three-year windows), and the base rate is itself close to the test's 5% level: persistence at the base rate is what the test would show if the yearly rejections were chance. Pairs share legs, so the intervals overstate the precision. Clegg (2014) reports the same pattern.
Source `results/v33_dev/anatomy.json` (persistence; the study's two-direction Engle–Granger test).

EXHIBIT 19

After formation, 60.8% of the traded pairs' half-lives exceed the rule's 25-day maximum

Spread half-life of each traded pair on its formation window and after formation, trading days (log scale)



Notes Median 17.1 days at formation, where selection admits 4 to 25, and 29.6 after formation. The dashed line is the rule's maximum and the curve marks equal half-lives; triangles mark half-lives beyond the top of the scale or no mean reversion. The time stop is at most 25 days, so a spread whose half-life has lengthened is usually closed before it reverts.
Source `results/v33_dev/anatomy.json` (traded pairs).

(101 signals turned away). A change in costs therefore changes which signals are filled. To separate the cost from the path, the headline run's 213 trades are also re-priced under each assumption. Re-pricing them for a $10M book turns their P&L slightly positive (+0.49%, Sharpe +0.10), because impact falls with participation; the full re-run at that size does not (−0.09%). Neither is distinguishable from zero, and a smaller book cannot turn a gross return that is itself indistinguishable from zero into an edge. The larger books are a different matter: participation is measured against full-day dollar volume while every fill is assumed in the closing auction, which trades only a fraction

TABLE 10

The v33 cost model against published references, and what changed from v32

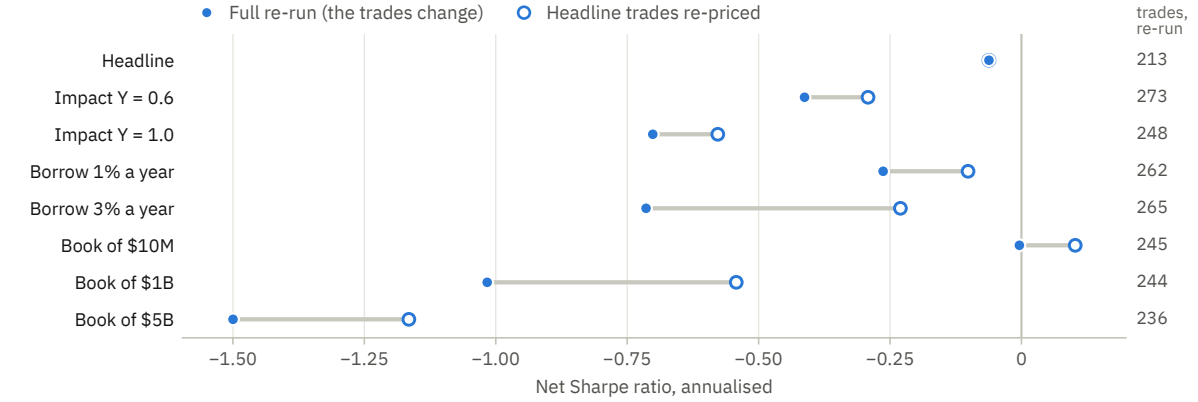
Component	v33 model	Reference	Change from v32
Commission	$0.002 per share, $0.50 minimum per order	Author's assumption, in the range of quoted agency rates	unchanged
Half-spread	Half the typical quoted spread by 20-day dollar ADV: 0.75 bp (≥ $500M) to 40 bp (< $5M)	Frazzini et al. (2018)	unchanged
Market impact	$Y \cdot \sigma_{daily} \cdot \sqrt{(\text{shares} / \text{ADV})}$, $Y = 0.3$, capped at 50 bp, on entry and exit	The 0.314 of Almgren et al. (2005) is their permanent-impact coefficient, not a trading cost; $Y = 0.3$ is at the low end of published estimates, consistent with impact in the closing auction (the execution-cost note gives the sources)	unchanged; stressed at $Y = 0.6$ and 1.0
Borrow	One fee of 0.38% a year on the short notional	D'Avolio (2002) on general-collateral fees; the fee is a median buy-side indicative fee, whose source the execution-cost note gives	replaces a 25 bp general-collateral fee; stressed at 1% and 3%
Financing	None beyond the borrow fee	v32's 50 bp prime-broker spread on the short half of gross double-counted the cost of shorting	folded into the single fee
Fill	Close of $t + 1$ for entry and exit, a market-on-close order; legs $1 : $β, shares fixed from the signal-day close	`docs/EXECUTION_COSTS.md`	unchanged; $t + 1$ open is a declared variant

Source Author's cost model for the v33 study profile; the references cited.

EXHIBIT 20

Under every declared cost and capacity scenario the full re-run loses money; each one is a new path, so the trades change

Net Sharpe ratio under each scenario fixed in advance: the whole backtest re-run (filled) and the headline run's 213 trades re-priced (open)



Notes Y is the square-root impact coefficient; the borrow scenarios charge every short 1% or 3% a year; the book sizes scale every position. Exhibit 21 extends the fixed-trade view to any cost multiple.
Source `results/v33_dev/cost_sensitivity.json` (`scripts/cost_sensitivity.py`); `results/v33_dev/anatomy.json` (impact, borrow and capacity re-pricing).

of it. At $1B and $5B the larger leg is 13.26% and 72.84% of a day's volume, orders that could not plausibly be filled at the close, and the 50 bp cap per execution binds there: those rows understate the cost and do not describe a feasible book under the fill model.

A second check re-sizes and re-costs the same trades under five other parameter sets of costs and position sizes (Exhibit 23; parameters in Table 20 in the appendix). The trades lose money under all five, and their gross P&L, net plus costs, is negative under all five too, from −0.32% to −0.08% of capital, against a positive gross in the headline run. The headline's positive gross depends on its sizing: weighted by size its trades earned more before costs than equal-weighted, so this check is about sizing as much as about costs.

TABLE 11

Declared scenarios: the full re-run, and the headline trades re-priced under the same assumption

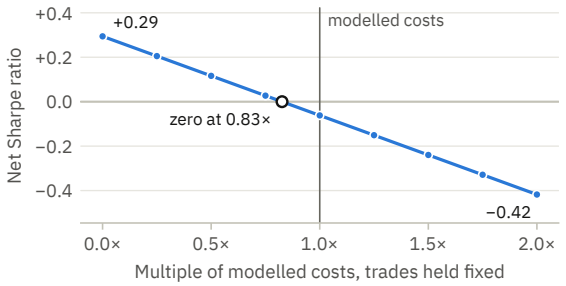
Scenario	Trades	Net return, %	Sharpe	Max drawdown, %	Cost, bp of notional	Larger leg, % ADV	Same trades: return, %	Same trades: Sharpe
Headline: Y = 0.3, borrow 0.38%, $100M	213	−0.39	−0.06	−4.05	16.9	1.61	−0.39	−0.06
Impact Y = 0.6	273	−2.66	−0.41	−4.96	27.8	1.41	−1.61	−0.29
Impact Y = 1.0	248	−4.38	−0.70	−6.30	39.1	1.46	−3.11	−0.58
Borrow 1% a year	262	−1.79	−0.26	−4.78	18.5	1.42	−0.60	−0.10
Borrow 3% a year	265	−4.61	−0.71	−7.06	24.5	1.42	−1.28	−0.23
Book of $10M	245	−0.09	0.00	−3.47	8.8	0.15	0.49	0.10
Book of $1B	244	−5.88	−1.02	−6.75	36.6	13.26	−2.92	−0.54
Book of $5B	236	−8.07	−1.50	−9.17	67.0	72.84	−6.13	−1.17

Notes The first seven columns are each scenario's full re-run; the last two hold the headline run's trades fixed and re-price them. Cost per round trip is in bp of traded gross notional, weighted by size.
Source `results/v33_dev/cost_sensitivity.json`; `results/v33_dev/anatomy.json`.

EXHIBIT 21

With the trades held fixed, net Sharpe falls from +0.29 at zero cost to −0.42 at twice the modelled costs

Net Sharpe ratio against a multiple of modelled costs, 0× to 2×, the headline trades held fixed

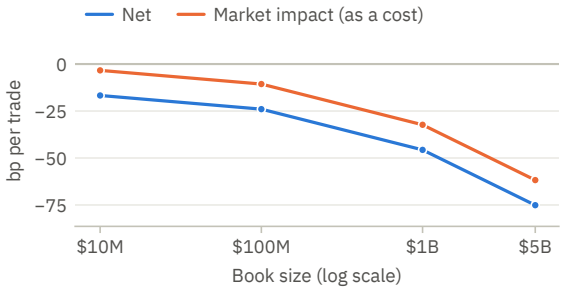


Notes Each day's net return is rebuilt as gross minus *k* times that day's modelled costs. The Sharpe ratio crosses zero at 0.83× costs (net P&L at 0.80×). Even at zero cost the Sharpe ratio is not significant (Newey–West p = 0.645).
Source The run's daily gross and cost series.

EXHIBIT 22

Larger books only lose more: impact grows from 3.4 bp a trade at $10M to 61.8 bp at $5B

Net and impact bp per trade when the headline trades are re-priced at other book sizes

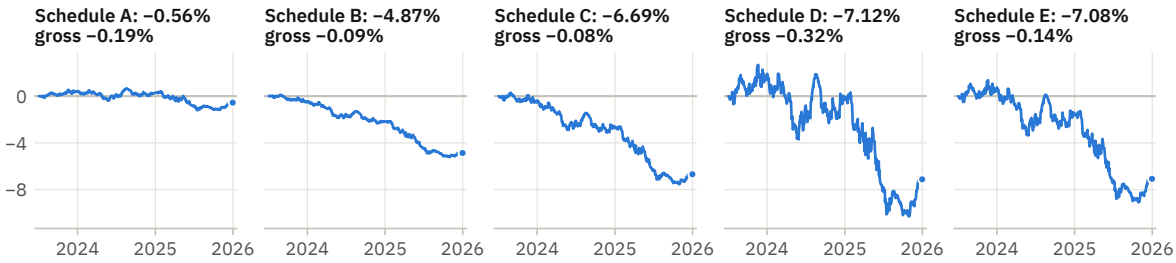


Notes Each execution is re-priced at the book's size: participation grows with the book, so square-root impact per dollar grows with it, up to the 50 bp cap per execution, which binds at the largest sizes, where the orders would also exceed what the closing auction can absorb. Net per trade: −16.7 bp at $10M, −24.0 bp at the headline $100M, −45.7 bp at $1B and −75.1 bp at $5B.
Source `results/v33_dev/anatomy.json` (capacity).

EXHIBIT 23

Re-sizing the same trades changes even the sign of their gross P&L: negative under all five other schedules, against $1.52M in the headline run

Cumulative net return, % of capital, when the main backtest's trades are re-sized and re-costed; schedules A–E in order of total cost, each panel titled with its net and gross return



Notes Schedules are parameter sets, not descriptions of any firm. Net returns run from −0.56% (schedule A, the lowest costs and smallest positions) to −7.12% (schedule D). The replay takes the main backtest's trades (206–213 of 213, under each schedule's daily entry limit) with the same entry and exit dates and daily paths, and re-sizes and re-costs them.
Source `results/v33_dev/fund_type_comparison.json`.

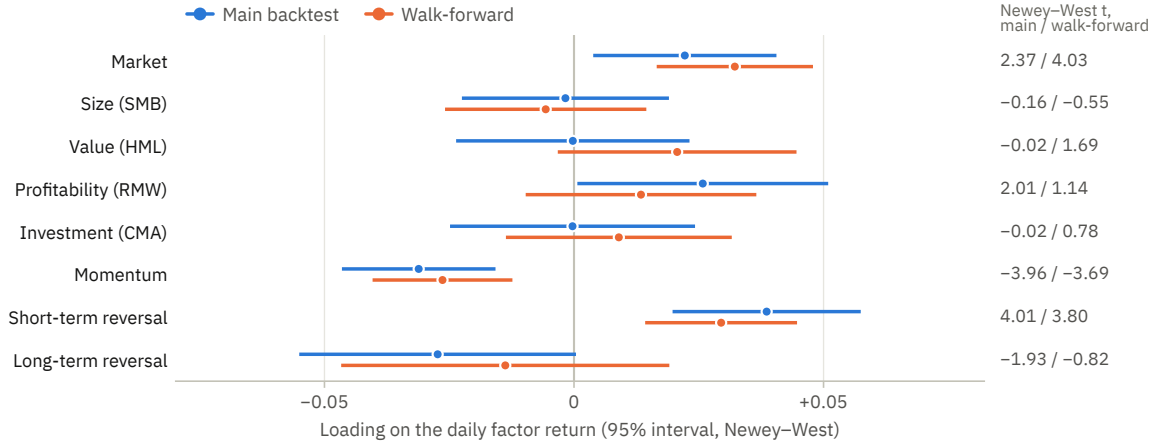
8 Risk and factor attribution

The book's factor exposures are the ones a convergence strategy built from hedge ratios below one should have. Against the five factors of Fama and French (2015), momentum (Carhart 1997) and short- and long-term rever-

EXHIBIT 24

Returns load on short-term reversal, as a convergence strategy's should, and leave no positive alpha

Loadings on eight daily factors with 95% intervals (Newey–West), main backtest and walk-forward

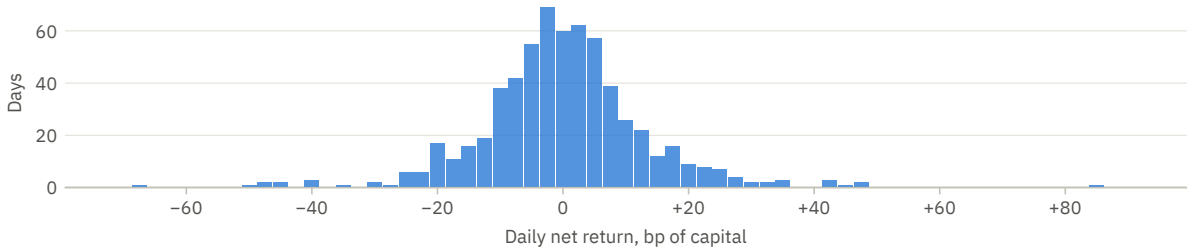


Notes Alpha a year: −0.51% (t = −0.38) in the main backtest and −2.59% (t = −2.25) in the walk-forward; R^2 0.158 and 0.148. Factor returns are truncated at 31 December 2025 in every run although the files reach into 2026; no strategy reads them.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, evidence block (factor attribution); factors from the Kenneth French data library.

EXHIBIT 25

Daily returns are fat-tailed (kurtosis 7.7)

Distribution of daily net returns, bp of capital; main backtest



Notes Skew 0.23; kurtosis is non-excess (3 for a normal distribution). Fat tails widen the Sharpe ratio's standard error, which the power figures of Section 5 account for.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance, daily net returns.

sal, with Newey–West errors, the main backtest loads on short-term reversal (t = 4.01), on the market (t = 2.37) and against momentum (t = −3.96) (Exhibit 24). The eight factors explain R^2 = 0.158 of its daily variance and leave an alpha of −0.51% a year (t = −0.38). The walk-forward loads the same way (short-term reversal t = 3.80) and leaves an alpha of −2.59% a year (t = −2.25): a loss, not an edge. Against an equal-weight sector-ETF proxy the main backtest's beta is 0.037 (t = 2.40): small, but not zero, as the hedge ratios below one imply (Exhibit 15).

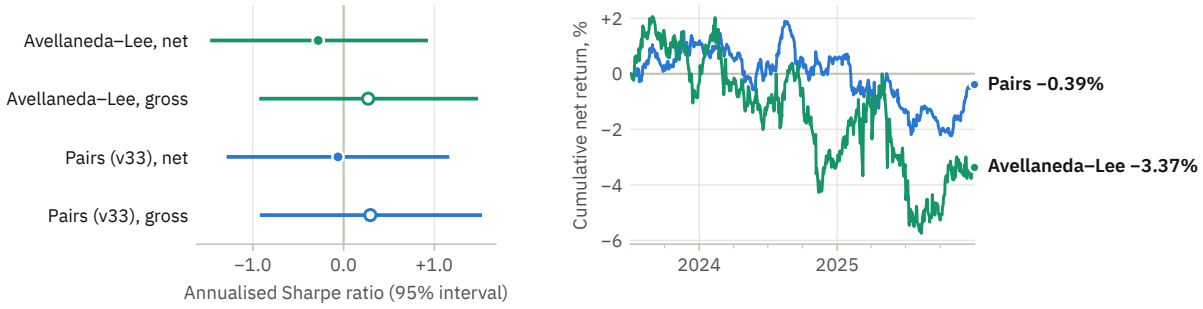
9 A second strategy family: residual reversion

A null on one rule says little about the idea. Avellaneda and Lee (2010) trade each stock's residual against its sector ETF when that residual, modelled as a mean-reverting process, is stretched; many small concurrent bets give it far more breadth than 213 pair trades, though over the same 628 days its Sharpe ratio is measured no more precisely (95% interval −1.47 to +0.93, against −1.29 to +1.16 for the pairs strategy). The parameters are the paper's: a 60-day regression on the sector ETF, an AR(1) fit of the cumulative residual,

EXHIBIT 26

The second family agrees: Avellaneda–Lee residual reversion has no detectable edge, gross (Sharpe +0.27) or net (−0.28)

Net and gross Sharpe ratios with 95% intervals (open markers: before costs), and cumulative net return, against the pairs strategy; same universe rule, fills and costs, July 2023 – December 2025



Notes Gross Sharpe 95% interval −0.93 to +1.48 (Newey–West p = 0.619). Costs were $6.09M against $2.71M of gross P&L at a turnover of 65.9× capital a year; the sector-ETF hedges are netted across positions and account for $0.76M of the costs.
Source `results/residual_dev/residual_statarb.json` (`scripts/run_residual_statarb.py`, run of commit 7442705); intervals by circular block bootstrap.

trades only when mean reversion is faster than 30 days, entry at $|s| > 1.25$, exits at −0.50 (long) and 0.75 (short). Fills, costs and the universe rule are the pairs strategy's: next-day closes, the same cost model, and the 500 most liquid point-in-time members, re-ranked every 21 trading days. The declared deviations from the paper are next-day fills, this cost model, 11 sector ETFs instead of 15, this universe, and a fixed position size (0.5% of equity) in place of a leverage target. Over the same period it made 10,916 trades, held 181 positions on average and returned −3.37% (Sharpe −0.28, 95% interval −1.47 to +0.93; Newey–West p = 0.598). Before costs it earned +5.35 bp a trade (t = 0.87, treating the overlapping trades as independent, which overstates its precision) against 11.30 bp of cost (Exhibit 26).

10 v32: the published record and the frozen arm

The v32 study was published on 18 October 2026 (commit 711b18c) with the same verdict: no detectable edge. v33 changes seven fields of its design for the reasons in Table 3. Two differences explain most of the contrast in Table 12. v32′s book was flat on 207 of 628 days, because its regime gate blocked new entries on 248 days; v33′s, without the gate, held positions almost every day near its 30% cap. And v32 searched any two of 300 names of a survivor list, where v33 searches same-industry pairs of 500 point-in-time members. Neither run's interval excludes zero. v32 is kept as the record of what was published and is not re-run with fixes; in the holdout it runs frozen as S3, reported with the same statistics as the tested arms but tested in no family, because its universe looks ahead into the test period (Section 11).

TABLE 12

The published v32 run beside the v33 run, development period

Measure	v32, published 18 October 2026	v33, this note
Main backtest, net Sharpe [95% interval]	−0.07 [−1.39, +1.23]	−0.06 [−1.29, +1.16]
Main backtest, net return	−0.30%	−0.39%
Main backtest, maximum drawdown	−3.00%	−4.05%
Walk-forward, net Sharpe [95% interval]	−0.33 [−1.49, +0.86]	−1.04 [−2.09, +0.05]
Walk-forward windows positive	3 of 11	4 of 11
Trades	161	213

Measure	v32, published 18 October 2026	v33, this note
Gross bp per trade, equal-weighted (t)	+11.4 (0.37)	−7.9 (−0.25)
Cost bp per trade, equal-weighted	12.5	16.1
Average gross exposure, % of capital	15.1	26.9
Days with no position	207	2
Pairs tested at the first formation	35,491	5,082
Gatev distance method, Sharpe	+0.07	−0.80
Percentile among random-pair portfolios	28	37

Notes Gross and cost per trade are equal-weighted across trades.
Source `results/v32_dev` (commit 711b18c); Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance.

11 The 2026 holdout test

Every design choice in this study was made on 2020–2025 data over 33 versions, so the development period cannot test those choices. The pre-registered confirmatory test runs on data that no part of the code or its caches has held: the 200 trading days from 2 January 2026 to 19 October 2026. That period has already ended, and the project was developed during it, from version 8 to version 33: the test is sealed, not prospective, and the designers' general knowledge of 2026 markets cannot be sealed. No daily price history of any security for dates after 31 December 2025 has been requested, and the fetch code refuses such a request until the registered commit unseals it. The protocol is filed first; each arm then runs once from the registered commit, every attempt is recorded, and the result is published whatever it is. At the date of this note the pre-registration is not yet filed; Table 13 gives the arms and hypotheses it is to fix.

TABLE 13
Arms and hypotheses of the pre-registration, to be filed before any 2026 daily price series is fetched

Label	Arm	Hypothesis
H1 · S1	v33 pairs, the design of this note	mean daily net return > 0
H2 · S2	Avellaneda–Lee residual reversion	mean daily net return > 0
S3	v32 as published, frozen	reported, not tested

Notes Test: one-sided Newey–West t on the daily net returns, Holm-adjusted across H1 and H2 at a familywise 5%. S3 is reported with the same statistics but tested in no family: its universe is ranked on market values inside the test period, so a rejection could reflect survivorship rather than an edge.

For each tested arm the null hypothesis is that the mean daily net return over the holdout is at most zero, against the alternative that it is positive. The test is the one-sided Newey–West t-test on every daily net return the arm's export records, with nothing dropped; the two confirmatory p-values are adjusted with the step-down procedure of Holm (1979) at a familywise level of 5%, and an arm shows an edge out of sample only if its adjusted p-value is below 0.05. The pre-registration also discloses the simulated size of this test at the holdout's length, which departs slightly from the nominal level. Whatever the outcome, each arm reports its net and gross return, its Sharpe ratio with standard error, the probabilistic Sharpe ratio, the block-bootstrap interval, the one-sided upper bound, its trades and exposure, the Ledoit–Wolf difference against the Gatev method, the random-pair percentile and, for S1 and S2, the survivorship gap at every formation or snapshot.

Power, stated in advance. About ten months of data can detect only a large Sharpe ratio. With returns shaped like the development period's (skewness 0.23, non-excess kurtosis 7.73, lag-1 autocorrelation −0.065), a test at Holm's first step (one-sided 2.5%, the level the better of the two arms must reach)

TABLE 14
Smallest annual Sharpe ratio the holdout detects with 80% power

Holdout length, trading days	Development shape, one-sided 2.5%	Development shape, one-sided 5%	Independent normal returns, one-sided 2.5%
126	3.74	3.31	3.98
200	2.96	2.62	3.15
252	2.63	2.33	2.81

Notes Six months, the registered calendar (2 January 2026 to 19 October 2026 on the exchange's holiday schedule) and a year. The development shape uses the asymptotic Sharpe variance for non-normal returns with lag-1 autocorrelation, as in Table 6; it is computed from development returns only.
Source Author's calculation from the daily net returns of the v33 development run.

with 80% power detects an annualised Sharpe ratio of about 2.96 over the registered 200 trading days, 3.15 for normal returns (Table 14); its power against a true Sharpe ratio of 1, 2 and 3 is 16%, 48% and 81%. The holdout is therefore a falsification check, not a proof of an edge: it can reject a large edge but cannot confirm a small one, and not rejecting the null is not evidence that the edge is zero, which is why each arm's upper bound is reported. Running v32 alone first would spend the holdout, so the three arms are frozen together and each runs once.

Inputs dated 2026. No strategy reads a 2026 daily price, but some inputs carry 2026 information, and the pre-registration lists each one. The v32 universe file ranks stocks on market values of 18 October 2026, which is why S3 is descriptive. The identity data (the CUSIP-to-ticker map, SEC industry codes, the listing venue) are as of October 2026: a ticker missing from the SEC's company list of that date has no industry, and in the holdout a name first held in a 2026 snapshot has one only if it is among the largest stocks of the v32 universe file, so both rules use 2026 information. The Kenneth French factor files and the FRED rates reach into 2026; every run truncates the factors at its own calendar's last date, and FRED is read only for the cash comparison. The vendor's responses carry the latest quote in their metadata, which no study code reads, and its adjusted price levels carry 2026 corporate actions, which the anchoring to N-PORT removes for S1 and S2.

12 Threats to validity

Each limitation, which way it biases the result, how large it is, and what addresses it (Table 15). Biases that flatter the strategy make the null more convincing, not less.

TABLE 15
Threats to validity of the v33 development run and the 2026 test

Threat	Direction of bias	Size	Mitigation
Survivorship is measured, not removed	↑ probably flatters	3.33% of index weight at the first formation and 1.08% at the last have no free price history: mostly names the vendor no longer serves (2.75 points at the first formation), largely acquired or delisted since, the rest failed fetches and failed data checks. Among the 500 heaviest members at the first formation, 19 have no history, including takeover targets and SVB Financial, which failed inside the walk-forward (Exhibit 2)	Point-in-time membership; the gap reported at every formation, split by fetch outcome, and at every formation and snapshot of the holdout
Design choices were made on the same data	↑ flatters	33 versions on 2020–2025; the deflated Sharpe's 32 trials are a lower bound; v33's changes were chosen after v32's results on this same period were seen; an exploratory v33 run was seen before the published one (Table 9)	The pre-registered 2026 holdout; the declared variant family; the trial registry

Threat	Direction of bias	Size	Mitigation
The 2026 test is sealed, not prospective	↑ may flatter the holdout	The test period ended on 19 October 2026, before the pre-registration is filed, and the project was developed during it. The code and caches hold no daily 2026 prices of the traded securities, but the designers' general knowledge of 2026 markets cannot be sealed	Every 2026 input received is listed (below, and in the pre-registration); designs, test and decision rule frozen before any 2026 daily price is fetched; one run per arm, every attempt on record
The path depends on the start date	· either way	Sizing is a share of equity and the 30% cap binds, so the trades taken depend on what the book already holds: on the 621 days they share, the main backtest and the walk-forward have Sharpe ratios of −0.09 and −0.82, a spread the bootstrap interval of one path does not include; the headline run is the more favourable of the two	Both paths reported, and the reason for each start date; the trades held fixed in the cost curve and the re-pricing; a distribution over staggered start dates is not yet run
Vendor prices carry corporate actions after the window	· either way	Split events are reported only inside the window, so early levels of names that split or paid dividends later are rescaled; anchoring moved 732 of 2,538 histories by more than 10%	Levels anchored to N-PORT for S1 and S2; returns, z-scores and hedge ratios are unaffected; a 2026 split's effect on the $5 screen is measured once 2026 split events can be fetched
Some inputs are dated 2026	· small, either way	Identity data as of October 2026 (two industry-map rules use 2026 information), factor files and FRED rates into 2026, quote metadata in the vendor's responses, and the v32 universe file	Rules disclosed in the pre-registration; factors truncated at each calendar's end and read by no strategy; FRED read only for the cash comparison; the quote metadata is not read
The frozen v32 arm's universe looks ahead into 2026	↑ flatters	Its membership is ranked by market value on 18 October 2026, inside the test period: a name that failed during 2026 is absent	S3 is descriptive: reported with the same statistics, tested in no family
Renamed securities can look delisted	· overstates the gap	A ticker change with a new CUSIP can read as a member without prices	Remapped through N-PORT's reported tickers, issuer successions and price checks against N-PORT's value per share
Borrow is one fee for every name	↑ flatters	Borrow was $130.7k of $1,907.4k of costs; specials and recalls are not modelled	0.38% median fee; 1% and 3% re-runs (Exhibit 20)
Fills are close-to-close	· either way	Larger leg 1.61% of full-day ADV on average, a larger share of the closing auction's volume; the closing auction is assumed, with no queue, auction-volume or intraday model	Impact stressed at $Y = 0.6$ and 1.0; $t + 1$ open fills a declared variant; the $1B and $5B books marked as beyond what the close can absorb
Cost parameters are published averages, not measurements	· either way	No free tick or quote data: spreads by ADV bucket and impact from published estimates	Declared stress scenarios; capacity re-pricing (Exhibit 20, Exhibit 22)
Cost scenarios change the trades	· either way	Each re-run is a new path: 213 to 273 trades	The headline trades re-priced beside each re-run, and the fixed-trade cost curve (Exhibit 21)
Hedge ratios below one leave market exposure	· adds beta	Mean hedge ratio 0.70; market loading t = 2.37 (Exhibit 24)	Eight-factor attribution reported; the market component of gross P&L separated (Exhibit 15)
The main backtest and the walk-forward overlap	· overstates the evidence	621 of 628 days in common, the same formations; with no model to refit they are one rule from two start dates	The holdout is the independent test
One period, one regime	· either way	30 months with three-month T-bills near 4.79%, and one path; returns are not split by volatility or dispersion regime, and Do and Faff (2010) find the strategy does best in prolonged turbulence	The regime gate is a declared variant; a breakdown by VIX and cross-sectional dispersion is not yet run
The random-pair control trades a different rule	· weakens the comparison	Random pairs run through the Gatev rule, not this engine, on committed capital	A control on the strategy's own engine is planned

Notes ↑ flatters the strategy; ↓ penalises it; · no single direction.

13 Forthcoming results

Each analysis below is specified and its code is committed, but its results are not yet among the committed data this note reads. When they are, the item leaves this table and appears in its section. None of them is reported here as a number.

TABLE 16
Forthcoming results

Analysis	What it will show	Awaits
Data-snooping tests	White's Reality Check, Hansen's SPA test, the Romano–Wolf stepdown, the model confidence set and the probability of backtest overfitting over the declared family (27-point grid, 11 toggles), whatever they show; the design is not re-chosen from them	The family's run on the development calendar, in progress
The 2026 holdout	Each arm's net Sharpe ratio, one-sided upper bound, Holm-adjusted p-value and decision (S1 and S2; S3 descriptive), and its cumulative return	The pre-registration, filed before any 2026 daily price is fetched, then a single run of each arm on 2026 data

14 Reproducibility

The research-note site presents the same results; Table 24 maps its exhibit numbers to this note's. Every number in this note is formatted by `scripts/build_paper.py` from `assets/study.json`, the file the research-note site's charts and checked numbers come from, and from the headline JSON that file was built from; the site's numbers are checked against the same JSON by `scripts/check_published_numbers.py`. Sentences whose wording depends on the data are asserted against it when the note is compiled. Prices come from Yahoo Finance's chart API, and a later fetch will not match the cached data bit for bit, because vendors revise history. The run manifest (Table 22) pins what was used, and the sha256 of every committed source the note reads is listed in Table 23; both are in the appendix.

The runbooks clear every `PAIRS_*` switch, fix the hash seed, refuse an environment that differs from `requirements.txt` and record the commit in every export. The execution-cost note, `docs/EXECUTION_COSTS.md`, compares the cost model with published impact estimates at each participation rate and gives the source of the borrow fee. The AR(1) variance of the Sharpe ratio and its source are documented in `pairs_trading/power.py`; the evidence layer is `pairs_trading/anatomy.py` and `pairs_trading/evidence.py`; the synthetic validation is `scripts/synthetic_validation.py`.

```
# the environment pinned in requirements.txt
python3.12 -m venv .venv
.venv/bin/pip install -r requirements.txt

# the development-period studies behind this note, in order:
# main, anatomy, costs, residual, variants, validation, publish
PYTHON=.venv/bin/python bash scripts/run_dev_studies.sh

# the site's data, the check of every marked number, and this paper
.venv/bin/python scripts/build_site_data.py           # results/v33_dev
.venv/bin/python scripts/check_published_numbers.py   # README.md, index.html
.venv/bin/python scripts/build_paper.py               # needs pip install -e ".[paper]"
.venv/bin/python scripts/build_paper.py --check

# the pre-registered 2026 test: once, from a full clone
# at the registered commit, after the pre-registration
PYTHON=.venv/bin/python bash scripts/run_holdout.sh
```

Disclosures

Hypothetical, backtested results on $100M of notional capital, not achieved by any account. Returns are net of the modelled costs described in Section 7 unless labelled gross, earn no interest on cash, and are shown beside

baselines run under the same costs. Hypothetical results have inherent limitations, including those listed in Section 12. This is a research project: not investment advice, not an offer of any product, and no claim of deployable performance. The cost schedules are parameter sets, not descriptions of any firm. Code under the MIT licence; IBM Plex typefaces under the SIL Open Font License 1.1.

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Appendix

TABLE 17

Specification of the v33 design (study profile v33, hash d54d21314c22)

Stage	Rule
Universe	Russell 3000 members as held by the iShares Russell 3000 ETF (IWV), from its quarterly SEC N-PORT filings: the latest snapshot whose period ends on or before each formation date, used from its period date, about two months before the filing was public (IWV's holdings stand in for the index membership FTSE Russell had already published). Holdings mapped to tickers with OpenFIGI and checked against N-PORT's own value per share. Daily prices 2020–2025: price history for 82.2% of the requested symbols; 14.2% no longer served by the vendor (HTTP 404) and 3.6% failed the fetch or the data checks; the fetch-coverage rule, which counts 404s and failed data checks as resolved, is met at 98.8%. Screens on point-in-time price (at least $5, anchored to N-PORT) and dollar volume (at least $10M a day), volatility and missing data; the 500 most liquid enter the pair search.
Selection	Pairs within one Fama–French 49 industry (SEC SIC codes); positive return correlation ≥ 0.25; two-direction Engle–Granger $p < 0.05$ (Bonferroni across directions); half-life 4–25 days on the cointegrating spread; quality score > 0.4. The whole selection re-run every 126 trading days on the trailing three years; open positions run off.
Spread	Two-state Kalman filter (slope and intercept) on log prices centred on their burn-in means, so β is a hedge ratio and signals are invariant to vendor back-adjustment.
Rule	One β-locked z-score against its mean and standard deviation over the last 30–60 bars, frozen at the signal. Entry $\lvert z\rvert \geq 1.8$; target once z reverts past −0.5 (long) or +0.5 (short); stop 1.5 further against; time stop max(10, min(⌊2.5 × half-life⌋, 25)) days. Gates: pair quality ≥ 0.55, 20-day correlation ≥ 0.20, β in [0.25, 4], a 30-day pause after three consecutive losing trades on a pair. No regime gate.
Book	$100M notional; 3–10% of equity per pair before risk scaling (2.2% to 10.0% as traded), scaled by the win rate of the last five closed trades and the volatility of recent daily returns; 30% gross cap on exposure at entry; at most 8 entries a day, ordered by signal strength × pair quality; no shared legs; daily mark-to-market; drawdown, daily-loss and exposure kill switch.
Costs	Per execution: $0.002 a share (minimum $0.50), half the quoted spread by 20-day dollar ADV, square-root impact ($Y = 0.3$, cap 50 bp); one 0.38% a year borrow fee on the short notional, charged over the days held.
Walk-forward	The same rule on the same formations from 14 February 2023, one continuous simulation cut into 63-day test windows; there is no model to refit.
Evidence	Probabilistic and deflated Sharpe, Newey–West t, Politis–White block bootstrap, power and upper bounds under a non-normal AR(1) variance, a Bonferroni Sharpe hurdle for 32 trials and two field-wide hurdles, Ledoit–Wolf Sharpe-difference test; eight-factor attribution; the anatomy of the result (shortfall, exit race, decay, persistence, half-lives, capacity, cash); Gatev distance method and random pairs under the same costs and t + 1 fills.

TABLE 18

Variants declared on 20 October 2026 for the data-snooping tests

Variant	What it changes
`ENTRY_Z`	grid values 1.50, 1.80, 2.10
`EXIT_Z`	grid values 0.25, 0.50, 0.75
`STOP_Z`	grid values 1.00, 1.50, 2.00
`regime_gate_on`	the v10/v11 gate fitted on now-test windows
`cusum_gate_on`	entry gate tried in earlier versions
`hurst_gate_on`	entry gate tried in earlier versions
`no_rolling_corr_floor`	decoupling filter (20-day correlation ≥ 0.20) removed
`ref_bars_40`	z-score reference window, shorter
`ref_bars_90`	z-score reference window, longer
`ranker_gbm`	the learned ranker (gradient-boosted trees) switched on
`open_fills`	t + 1 open instead of t + 1 close fills
`all_pairs`	no same-industry restriction
`universe_300`	v32's 300 most liquid names
`reform_every_63`	quarterly instead of half-yearly re-formation

Notes The threshold grid covers the three hand-set trading thresholds, which were chosen across versions on this same data. Each toggle changes one setting of the v33 design.
Source The declaration of the variant family, `results/variant_family/declaration.json`, committed on 20 October 2026.

TABLE 19

Walk-forward windows: calendar slices of one continuous simulation

Window	Test from	Test to	Days	Net return, %	Sharpe	Trades opened	Win rate, %	Max drawdown, %
1	14 Feb 2023	15 May 2023	63	−1.67	−3.02	28	35.7	−1.87
2	16 May 2023	15 Aug 2023	63	−0.58	−1.45	22	50.0	−0.68
3	16 Aug 2023	13 Nov 2023	63	−1.64	−3.32	22	27.3	−1.87
4	14 Nov 2023	14 Feb 2024	63	+0.14	+0.33	20	45.0	−0.91
5	15 Feb 2024	15 May 2024	63	−1.65	−3.47	23	39.1	−1.85
6	16 May 2024	15 Aug 2024	63	+1.10	+1.89	21	38.1	−0.73
7	16 Aug 2024	13 Nov 2024	63	+0.58	+1.24	22	59.1	−0.55
8	14 Nov 2024	18 Feb 2025	63	−0.76	−1.42	18	61.1	−1.19
9	19 Feb 2025	19 May 2025	63	+0.73	+0.80	31	41.9	−1.75
10	20 May 2025	19 Aug 2025	63	−1.28	−2.62	24	45.8	−1.52
11	20 Aug 2025	19 Dec 2025	86	−1.34	−2.42	25	48.0	−1.73

Notes Windows are numbered in calendar order. Positions carry across window boundaries. Sharpe ratios are annualised. The last window runs to the walk-forward's last day.
Source Author's backtest, v33 development run (commit 7442705); prices from Yahoo Finance.

TABLE 20

Cost schedules A–E: parameters and results of the replay

	Comm., bp	Half-spread, bp	Impact cap, bp	Borrow, % a year	Financing, % a year	Position, % of equity	Gross, %	Costs, %	Net return, %	Sharpe	Max DD, %	Peak gross, × capital	Trades replayed
A	1.0	1.5	4	0.10	0.00	2–5	−0.19	0.37	−0.56	−0.26	−1.85	0.21	210
B	0.7	15.0	80	6.00	3.00	2–5	−0.09	4.78	−4.87	−2.33	−5.33	0.21	206
C	5.0	10.0	50	2.00	1.50	4–10	−0.08	6.61	−6.69	−1.60	−7.81	0.42	212
D	1.0	2.0	12	0.60	0.50	12–30	−0.32	6.80	−7.12	−0.55	−12.63	1.25	213
E	1.5	4.0	20	1.00	1.00	8–20	−0.14	6.94	−7.08	−0.84	−10.30	0.83	213

Notes Gross is the replay's net P&L plus its costs, % of capital. Each schedule applies its own daily entry limit, so it replays up to all of the main backtest's trades.
Source Author's replay of the v33 trades under schedules A–E; returns, Sharpe ratios and drawdowns as in the run's headline results.

TABLE 21

Selected versions of the study

Version	Change
v9–v12	First look-ahead fixes (P&L, pair selection, Kalman β drift).
v16–v19	Kalman spread fixes, β-weighted P&L, PCA residual cointegration, dynamic hold.
v24–v26.1	Ranker trained and wired in (ranking only); controlled ablation; a label bug that had disabled training.
v28–v29	Significance layer and baselines; $t + 1$ entry fills.
v30	Leakage audit: survivorship confirmed, selection-clean window tagging.
v31–v31.1	Pre-launch audit: inverted stop-loss, same-bar exits, future re-selections in the walk-forward; results withdrawn.
v32	Full audit and rebuild of the engine, Kalman model, accounting, walk-forward, costs, ranker, benchmarks and outputs; synthetic known-answer validation; study run of 18 October 2026: no detectable edge. Kept as the published record, and as the frozen descriptive arm of the holdout.
v33	The design to be pre-registered: point-in-time universe from N-PORT, same-industry pairs, full re-formation, regime gate and ranker off, one borrow fee; the evidence layer, a second strategy family, the trial registry and the sealed 2026 holdout. Development run of 21 October 2026: no detectable edge.

Notes Versions that only withdrew or corrected earlier results are listed with the fixes they made; `docs/PROGRESS.md` has the full record.

TABLE 22

Run manifest

Item	Value
Commit	`744270556ba6 (clean)`
Run	`2026-10-21 06:19 UTC; results in results/v33_dev/`
Design	`profile v33 d54d21314c22, calendar dev 0bcc31796bad`
Price cache	`sha256 3a9a93cc2ea9352e7e43319307ec9039cf0bd4f35c06939dc5a956c619c0a7be`
Headline JSON	`results/v33_dev/headline_metrics.json, sha256 b09f239032db62196602447e8d1d780286b5c2674e8e85d02062beab946d0fb1`
Site data	`assets/study.json, sha256 d8e8fca7332c56e51314d271b8244d2a99ced64c8d8dde04597f6592daefc5ea`
Universe	`IWV N-PORT snapshots (reference/universe/); price history for 82.2% of the requested symbols (2,538), 14.2% no longer served (HTTP 404); fetch-coverage rule met at 98.8%`
Fetch manifest	`results/v33_dev/price_fetch_manifest.json`
Seed	`PAIRS_SEED=42, PYTHONHASHSEED=0`
Stack	`python 3.12.3 · numpy 1.26.4 · pandas 2.2.3 · scipy 1.15.2 · statsmodels 0.14.4`

TABLE 23

Committed sources read by this note

File	sha256
reference/universe/iwv_2022-12-30.csv	72768e4abbef6d0ec94e249b05fc7b6681822fe4c5511d212463d67ad0688f63
reference/universe/iwv_2023-06-30.csv	80be0c0e308e7634e7ffba4444a2185532114539e984ffd63ec3fca997fe0e65
reference/universe/iwv_2023-12-31.csv	d1c115e4abdb710cd592ef11d2f7d91318675be51fcfa166e3a7f63c5cd6c4e5
reference/universe/iwv_2024-06-30.csv	95837ee080ec2d627747aa44ed37b5f7d48c143540e8ce3f385ae09e6fe5a82a
reference/universe/iwv_2024-12-31.csv	acdd6cecd2fbd94082e5226aff910f33efba53cafb3039e60310057d12f888df
reference/universe/iwv_2025-06-30.csv	767b8602838b419c9e7ebab5f76a113938e28df0c433c00064d7ebc9932304f0
results/residual_dev/MANIFEST.json	1a89276294d27e0b9c6dd1200408515340384096 2da91659c3ace30df8a4e0ee
results/residual_dev/residual_statarb.json	7bc4562bf017ce2bb31763ea97effa7e98049abbdd74a27faf452147aed2b222
results/trial_registry.jsonl	1d3d9219aaab973f490ead6fa8580b43a71644bd2d33bf3b4981dedecb40e990
results/v32_dev/headline_metrics.json	934715640095540 07e8bff1f6a67a02f7f2db96b99d76ad8bd3e5a57b5183b0e
results/v32_dev/pairs_trading_results_2023_2025_unbiased.json	0d2e376fdbac5de777aa33187938d9e0d23ef76207f4cdc92de93ce260f9e876
results/v33_dev/anatomy.json	f181726d79fb3150638aa29642585dcd72f0078a5fc2f16f2fd031d80b2beb0d
results/v33_dev/cost_sensitivity.json	3915a9bd2e94aedc351af47869d524bcf1c0e86e044446094b0ecfeccaded1d2
results/v33_dev/fund_type_comparison.json	e1ecc11df56a0049435a168ab0c2ea840a613f42491bafc882e017bde70e8205
results/v33_dev/headline_metrics.json	b09f239032db62196602447e8d1d780286b5c2674e8e85d02062beab946d0fb1
results/v33_dev/pairs_trading_results_2023_2025_unbiased.json	0209861c37c3bb97bd9a2f79723490956a4a35af20462a396ec5493622f06b21
results/v33_dev/price_fetch_manifest.json	b00c6c31d28daee14f46797c3bc3e9f2239c1120e9fdf31bdece2fa94c3d4fd8
results/validation/synthetic_validation.json	1dd037fdb4d53d4946b3d5182226c45f0cd1f29171e85451ec7f7dcb2ba4485b
results/variant_family/declaration.json	028b1a29935565931fbe191ac578d8f2756d36b1ddc1d7ecb4833b8f7ca66add
results/v32_dev/universe_meta.json	af2bc3d41d0b2cee0381c6479baa144d1a743cde7f78dc266de3bddac20116b5
reference/universe/index.json	2afd0cb6a7a7b79e08e6d45a5fc78f7ac6d43903c3cafbde1a042bf048883c3c
index.html	3fcc3f0d796b2c8854ecabd882a0950b44f6af0458b149890b6d2b23bd3f923c

Notes The PDF is compiled with a fixed timestamp, the note's date, so a rebuild of unchanged inputs is byte-identical; the build manifest records the sha256 of every compile input and of the PDF.

TABLE 24

Exhibits of the research-note site and where they appear in this note

Site	Title on the site	This note
Exhibit 1	Net Sharpe ratio of each pre-registered arm on the 2026 holdout	forthcoming (Table 16)
Exhibit 2	Net of costs the book ends where it started, −0.39% after two and a half years, with the 30% gross cap close to binding throughout	Exhibit 6
Exhibit 3	The same rule started in February 2023: 4 of 11 quarterly windows made money and the stitched path ends at −6.24%	Exhibit 7
Exhibit 4	No month carries the result: 13 of 30 months were positive and the best, August 2023, earned +1.13%	Exhibit 8
Exhibit 5	Half-year Sharpe ratios range from −2.5 to +1.2, and no window's band clears zero	Exhibit 9
Exhibit 6	Every bootstrap interval spans zero, gross or net, in both families; the main backtest excludes Sharpe ratios above 0.92, and the walk-forward's one-sided bound is below zero	Exhibit 10
Exhibit 7	The strategy sits at the 37th percentile of 200 random-pair portfolios	Exhibit 11
Exhibit 8	Deflated for 32 trials, the main backtest's Sharpe ratio is far from significant (deflated Sharpe ratio 1.4%); the data-snooping tests are running	Exhibit 12
Exhibit 9	From paper P&L to net: the next-day fill gives up part of the paper P&L, and costs exceed what the trades earned before them	Exhibit 14
Exhibit 10	Targets and stops offset each other; long-spread trades made $1.04M and short-spread trades lost $1.43M, consistent with hedge ratios below one	Exhibit 15
Exhibit 11	After formation, spreads hit their targets about as often as a random walk would, not as the selection implies	Exhibit 16
Exhibit 12	The rule's edge on its own pairs is in-sample only: +64.7 bp a trade on the selection data, −12.6 bp after formation	Exhibit 17
Exhibit 13	Cointegration persists only at the base rate: a pair cointegrated one year is cointegrated the next about as often as any correlated pair	Exhibit 18
Exhibit 14	After formation, 60.8% of the traded pairs' half-lives exceed the rule's 25-day maximum	Exhibit 19
Exhibit 15	Under every declared cost and capacity scenario the full re-run loses money; each one is a new path, so the trades change	Exhibit 20
Exhibit 16	With the trades held fixed, net Sharpe falls from +0.29 at zero cost to −0.42 at twice the modelled costs	Exhibit 21
Exhibit 17	Larger books only lose more: impact grows from 3.4 bp a trade at $10M to 61.8 bp at $5B	Exhibit 22
Exhibit 18	Re-sizing the same trades changes even the sign of their gross P&L: negative under all 5 other schedules, against +$1.52M in the headline run	Exhibit 23
Exhibit 19	Returns load on short-term reversal, as a convergence strategy's should, and leave no positive alpha	Exhibit 24
Exhibit 20	Daily returns are fat-tailed (kurtosis 7.7)	Exhibit 25
Exhibit 21	From 5,082 pairs tested at the first formation, chance alone could account for up to about half the “cointegrated” ones	Exhibit 13
Exhibit 22	The second family agrees: Avellaneda–Lee residual reversion has no detectable edge, gross (Sharpe +0.27) or net (−0.28)	Exhibit 26
Exhibit 23	The pipeline, and what data each stage may see	Exhibit 3
Exhibit 24	Data splits: formation and re-formations, walk-forward, main backtest and the sealed holdout	Exhibit 4
Exhibit 25	The survivorship gap, measured: index weight without price history falls from 3.33% at the first formation to 1.08% at the last	Exhibit 2
Exhibit 26	On synthetic markets the engine finds a planted edge (+82.6 bp a trade) and none where there is none (+1.7 bp, t = 0.65)	Exhibit 5

Notes The site orders its exhibits by its own sections. Titles are the site’s; where this note words a finding differently, this note’s wording applies. Exhibits of this note without a site counterpart are not listed.